



US012414340B2

(12) **United States Patent**
Lee et al.

(10) **Patent No.:** **US 12,414,340 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **SEMICONDUCTOR STRUCTURE INCLUDING NANOSHEET CHANNEL STRUCTURE AND METHOD FOR FORMING THE SAME**

(71) Applicant: **TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY LTD.**, Hsinchu (TW)

(72) Inventors: **Hsin-Yi Lee**, Hsinchu (TW); **Weng Chang**, Hsin-Chu (TW); **Chi On Chui**, Hsinchu (TW)

(73) Assignee: **TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY LTD.**, Hsinchu (TW)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 10 days.

(21) Appl. No.: **17/388,465**

(22) Filed: **Jul. 29, 2021**

(65) **Prior Publication Data**
US 2023/0034854 A1 Feb. 2, 2023

(51) **Int. Cl.**
H10D 62/10 (2025.01)
H10D 30/01 (2025.01)
H10D 30/67 (2025.01)

(52) **U.S. Cl.**
CPC **H10D 62/121** (2025.01); **H10D 30/031** (2025.01); **H10D 30/6735** (2025.01); **H10D 30/6757** (2025.01)

(58) **Field of Classification Search**
CPC H01L 29/0673; H01L 29/42392; H01L 29/66742; H01L 29/78696; H10D 62/121; H10D 30/6757; H10D 30/6735; H10D 30/031

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

9,209,247 B2 12/2015 Colinge et al.
9,236,267 B2 1/2016 De et al.
9,412,817 B2 8/2016 Yang et al.
9,412,828 B2 8/2016 Ching et al.
9,472,618 B2 10/2016 Oxland
9,502,265 B1 11/2016 Jiang et al.
9,520,482 B1 12/2016 Chang et al.
9,536,738 B2 1/2017 Huang et al.
9,576,814 B2 2/2017 Wu et al.
9,608,116 B2 3/2017 Ching et al.

(Continued)

Primary Examiner — Norman D Richards

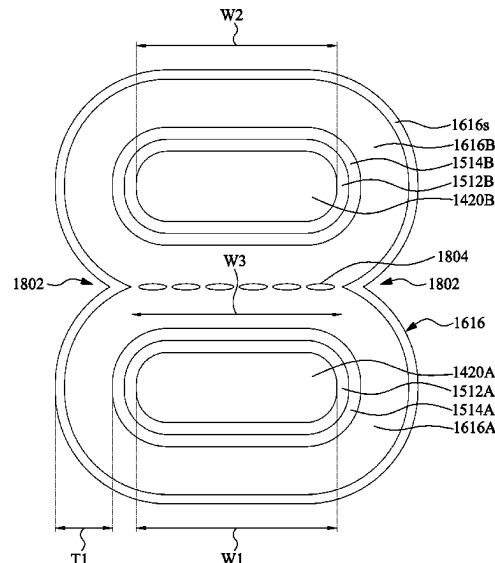
Assistant Examiner — Sean Ayers Winters

(74) *Attorney, Agent, or Firm* — WPAT LAW; Anthony King

(57) **ABSTRACT**

The present disclosure provides a semiconductor structure and a method for forming the same. The semiconductor structure includes a first nanosheet channel structure and a first high-k dielectric layer surrounding the first nanosheet channel structure. In addition, the semiconductor structure includes a second nanosheet channel structure disposed above and substantially parallel to the first nanosheet channel structure, and a second high-k dielectric layer surrounding the second nanosheet channel structure. The semiconductor structure further includes a work function adjustment layer including silicon and disposed between the first high-k dielectric layer and the second high-k dielectric layer. The first high-k dielectric layer and the second high-k dielectric layer are separated by the work function adjustment layer.

20 Claims, 29 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2005/0006703	A1 *	1/2005	Miscione	H01L 27/12 438/149
2015/0187892	A1 *	7/2015	Yin	H01L 29/7846 438/296
2016/0181107	A1 *	6/2016	Shin	H01L 21/28132 438/592
2016/0315088	A1 *	10/2016	Kang	H01L 29/513
2019/0081152	A1 *	3/2019	Suh	H01L 29/66545
2019/0348541	A1 *	11/2019	Moroz	H01L 21/30604
2019/0378906	A1 *	12/2019	Loubet	H01L 29/66742
2020/0035567	A1 *	1/2020	Chanemougame ...	H01L 29/401
2020/0075720	A1 *	3/2020	Cheng	H01L 29/78696
2020/0098756	A1 *	3/2020	Lilak	H10D 62/83
2020/0111885	A1 *	4/2020	Lin	H01L 29/66568
2020/0152746	A1 *	5/2020	Tsau	H01L 23/53266
2021/0098589	A1 *	4/2021	Cheng	H01L 21/823412
2021/0217654	A1 *	7/2021	Xie	H01L 21/22

* cited by examiner

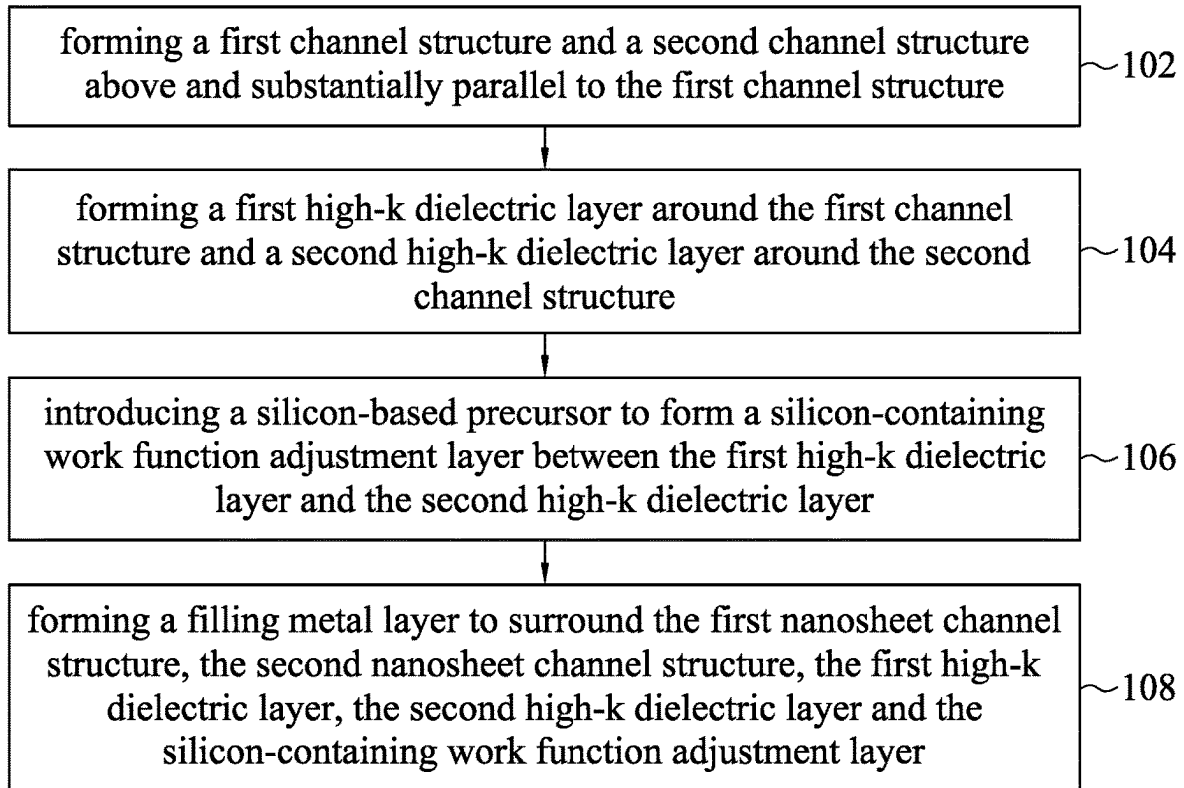
10

Fig. 1

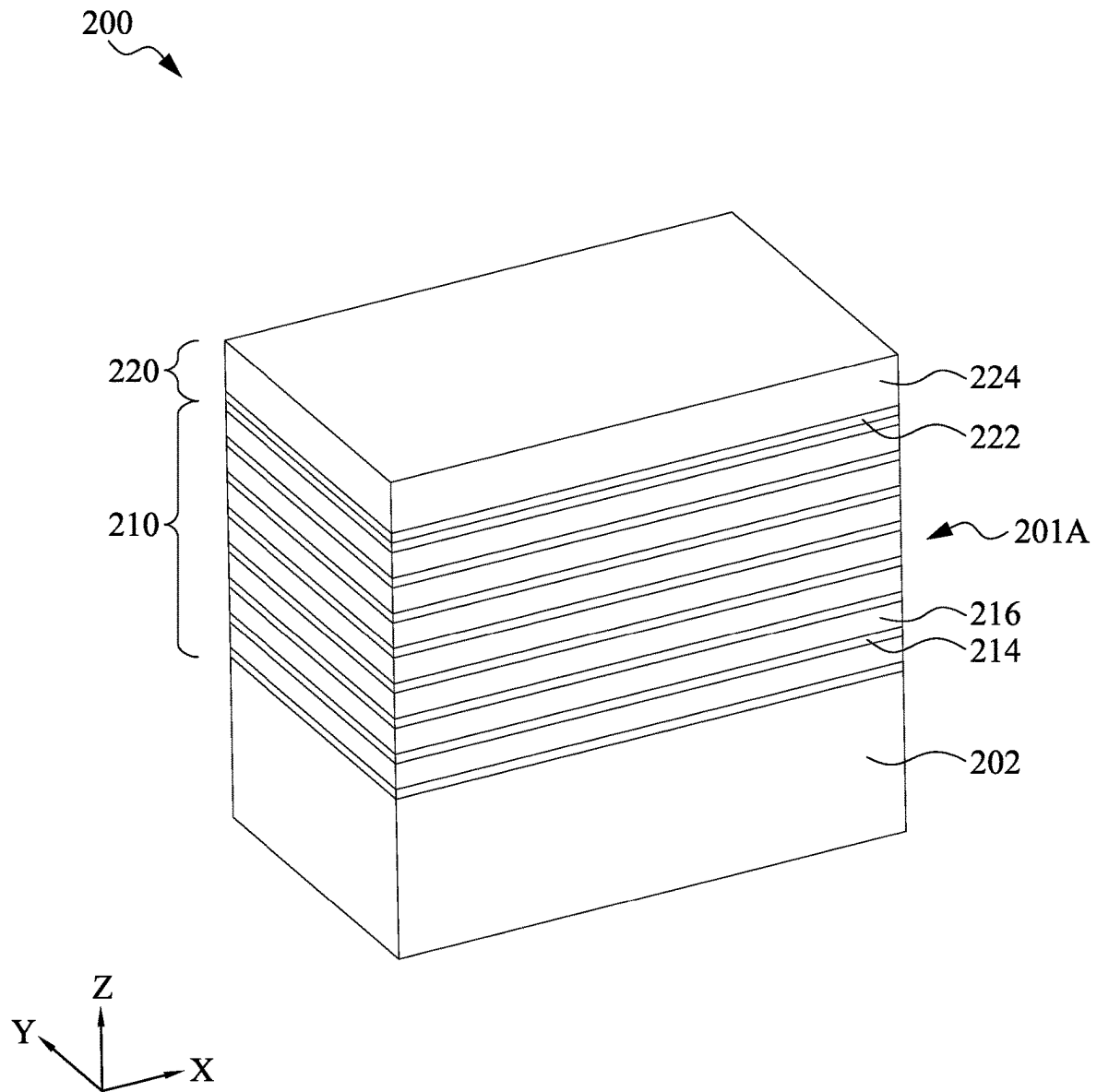


Fig. 2

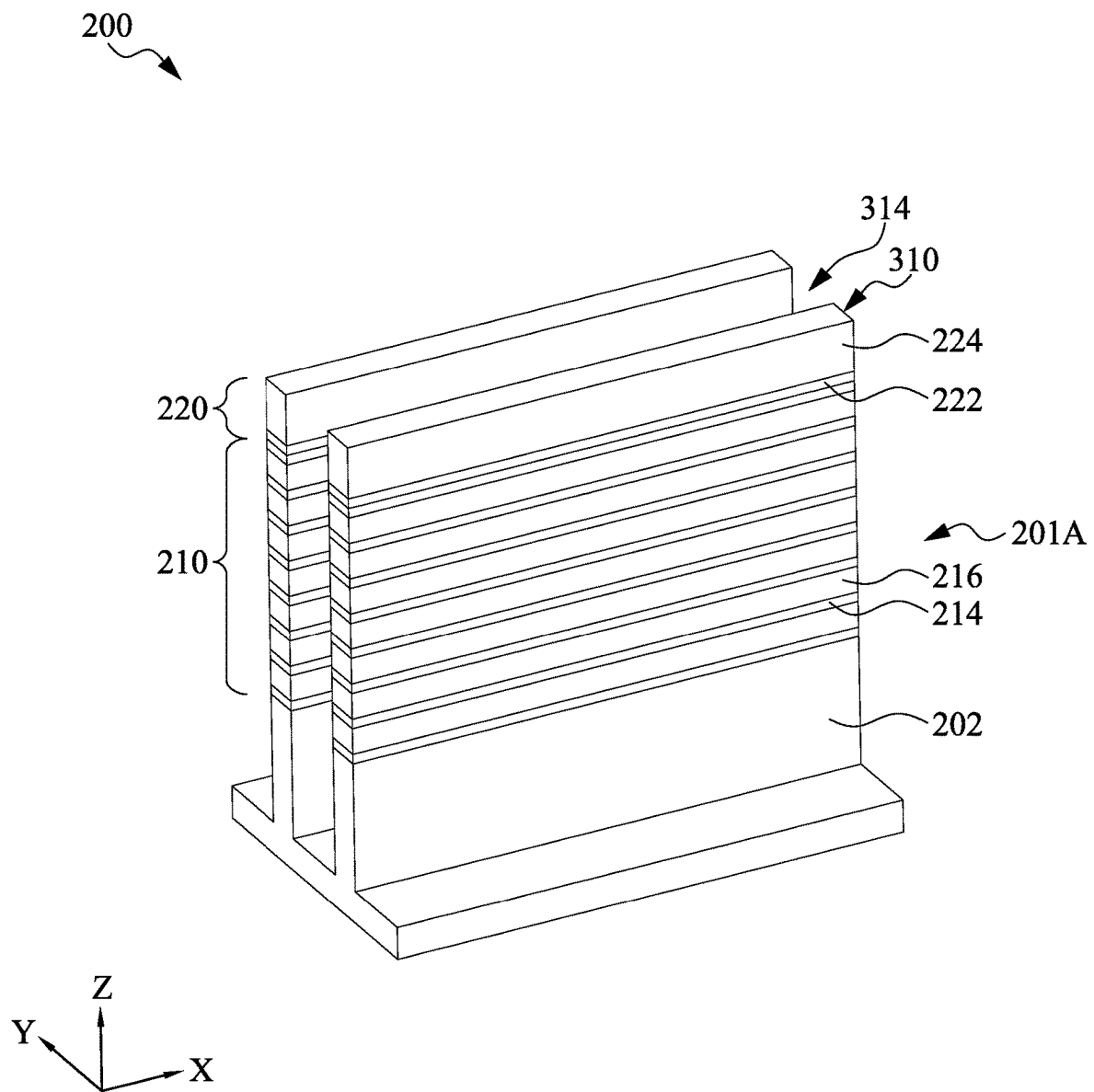


Fig. 3

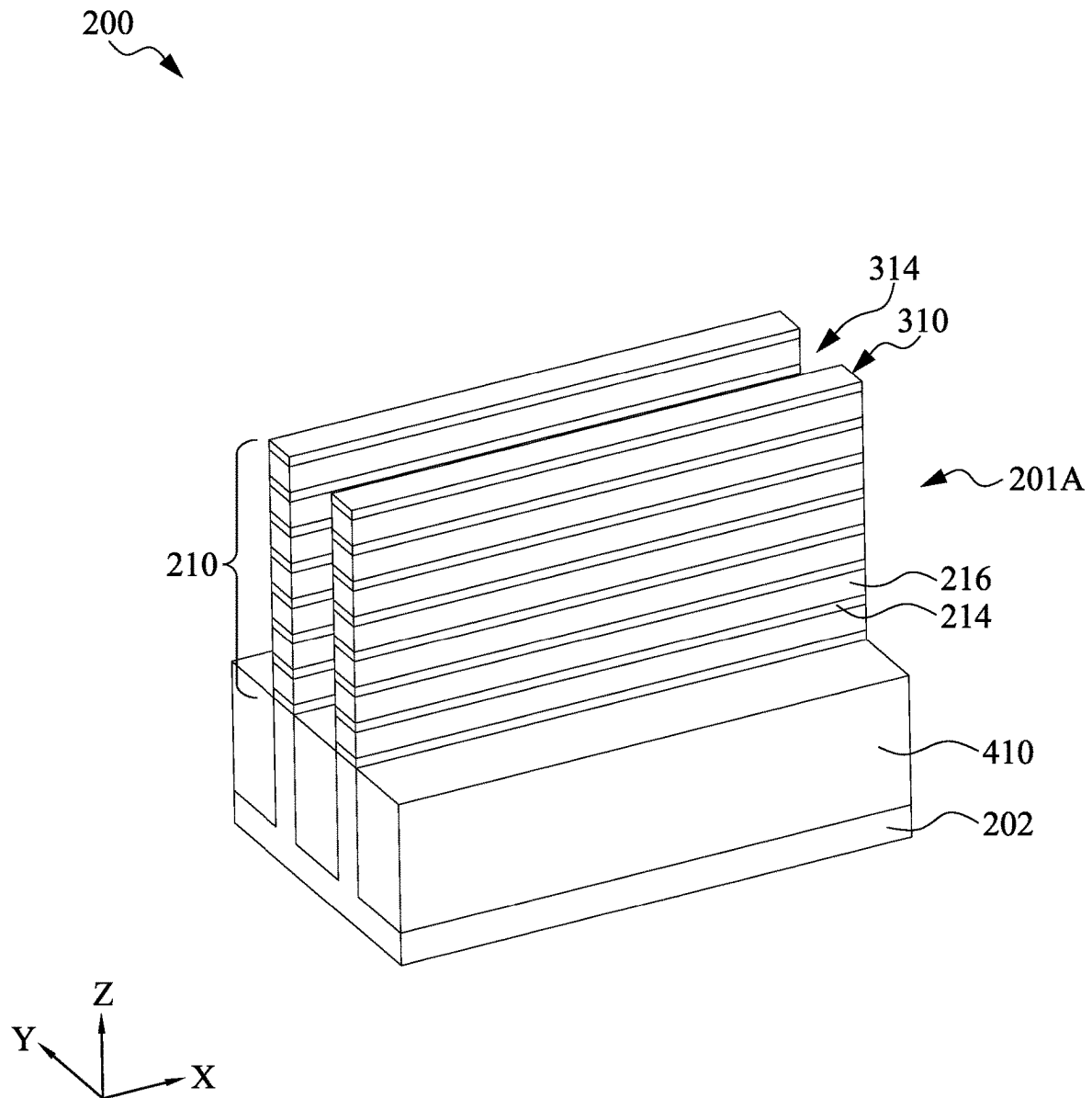


Fig. 4

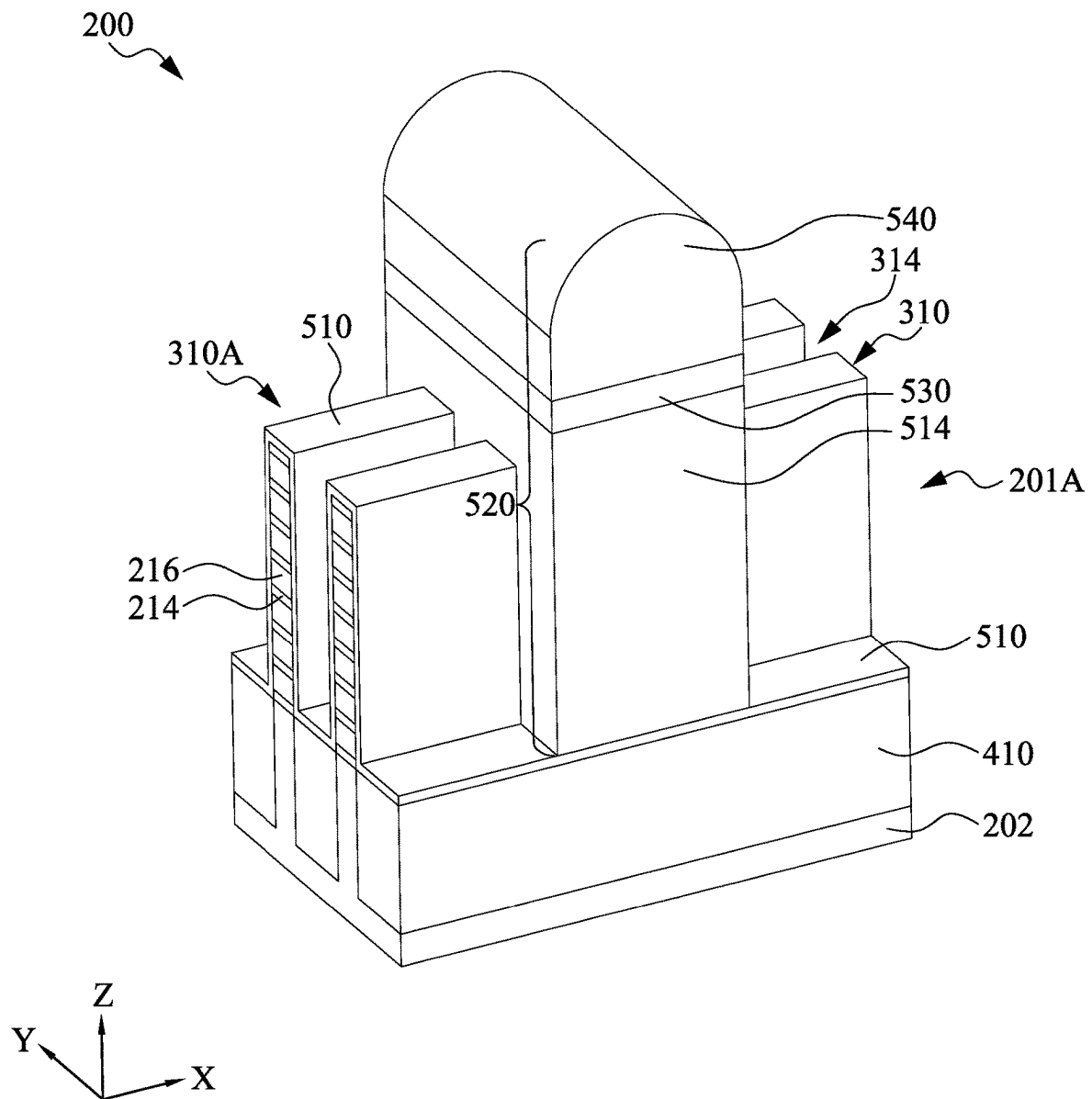


Fig. 5

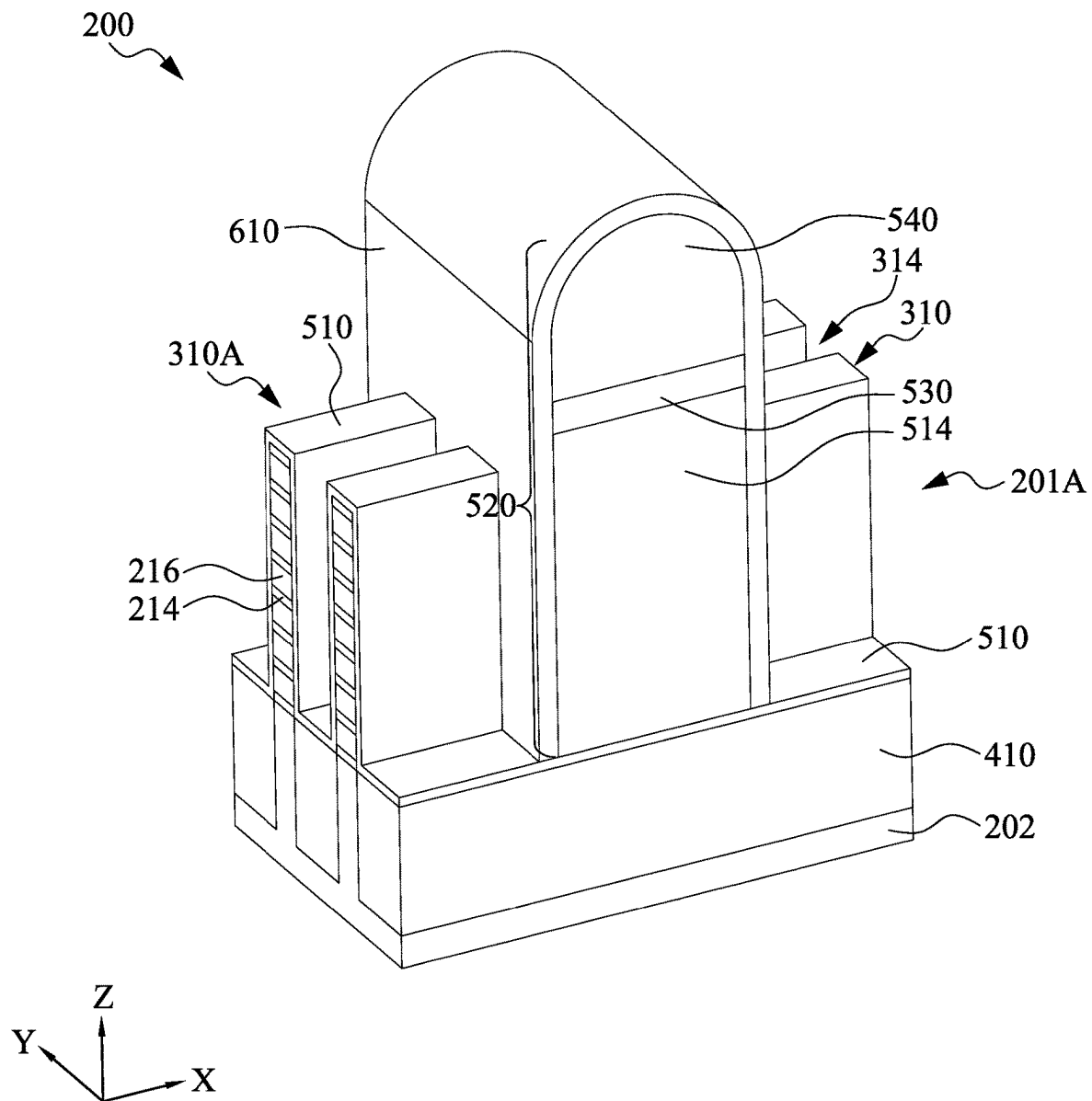


Fig. 6

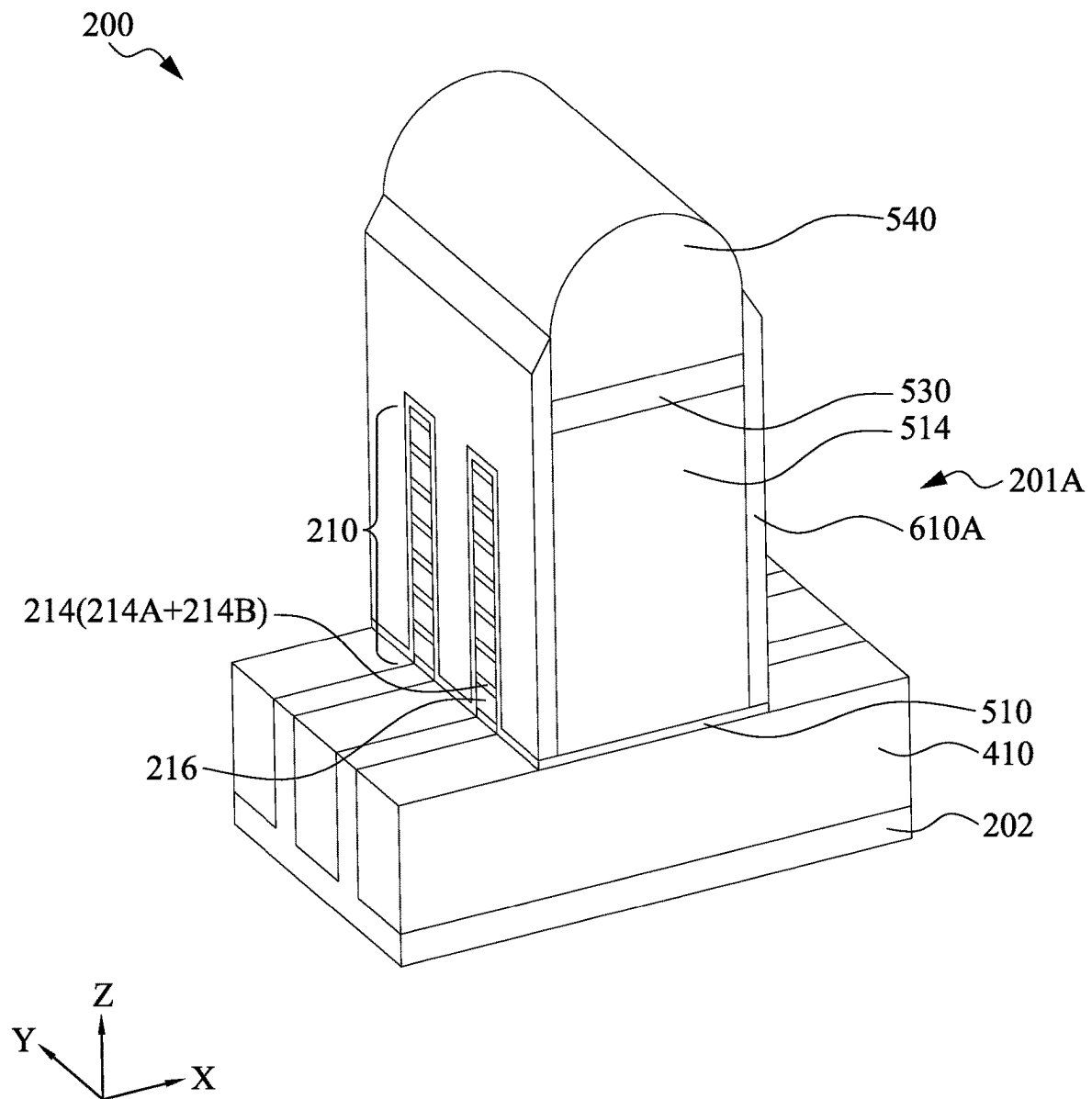


Fig. 7

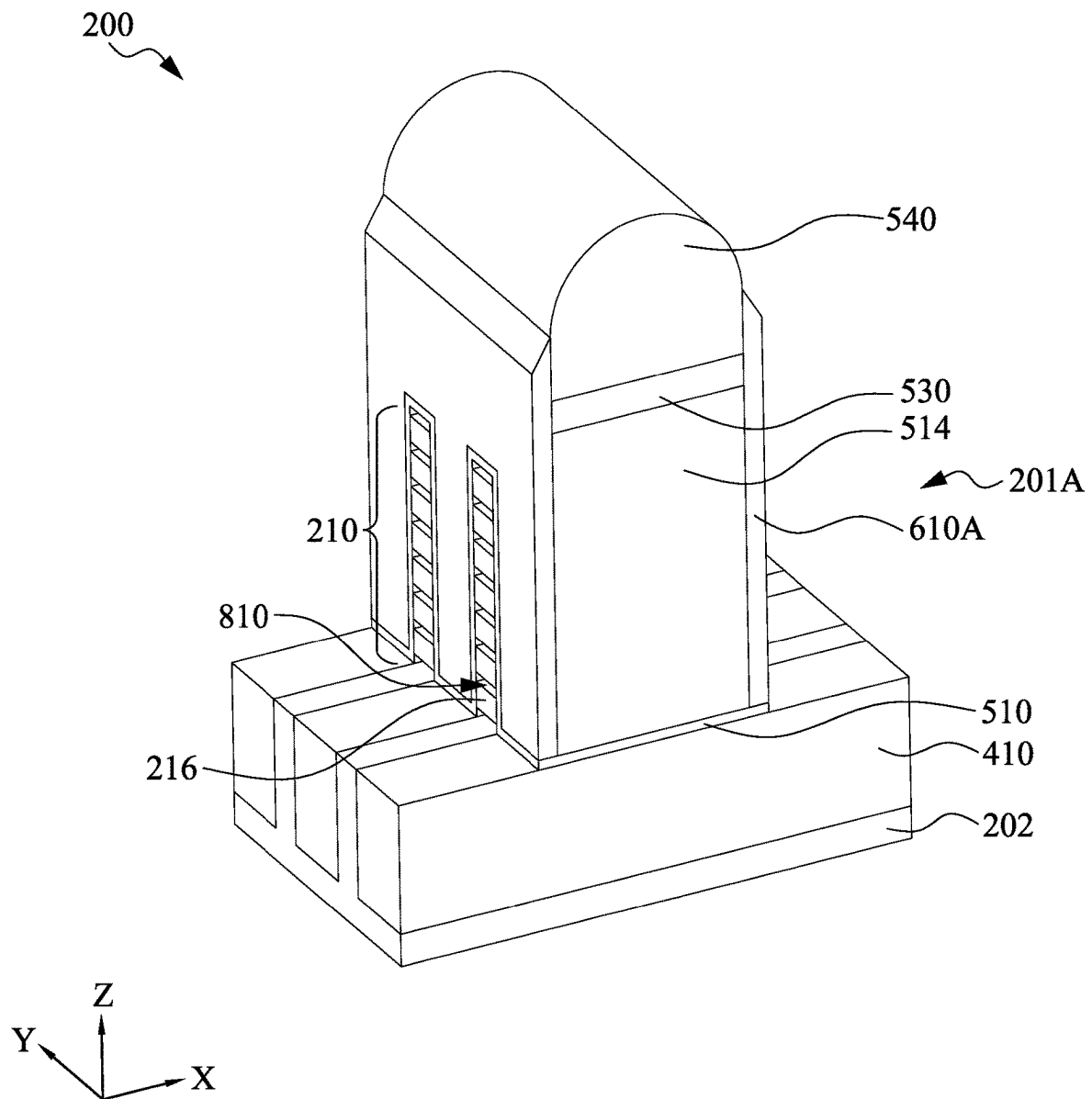


Fig. 8

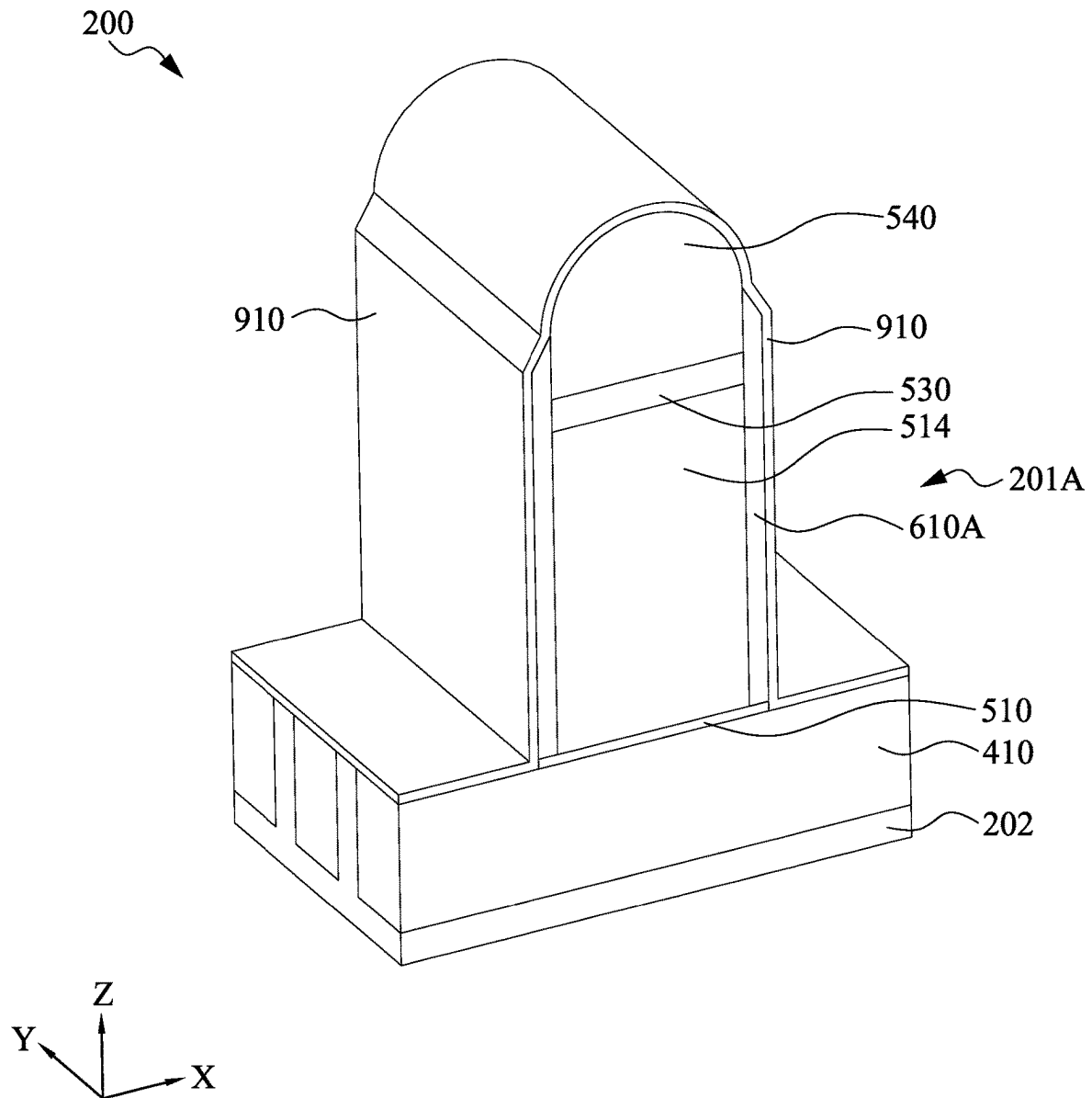


Fig. 9

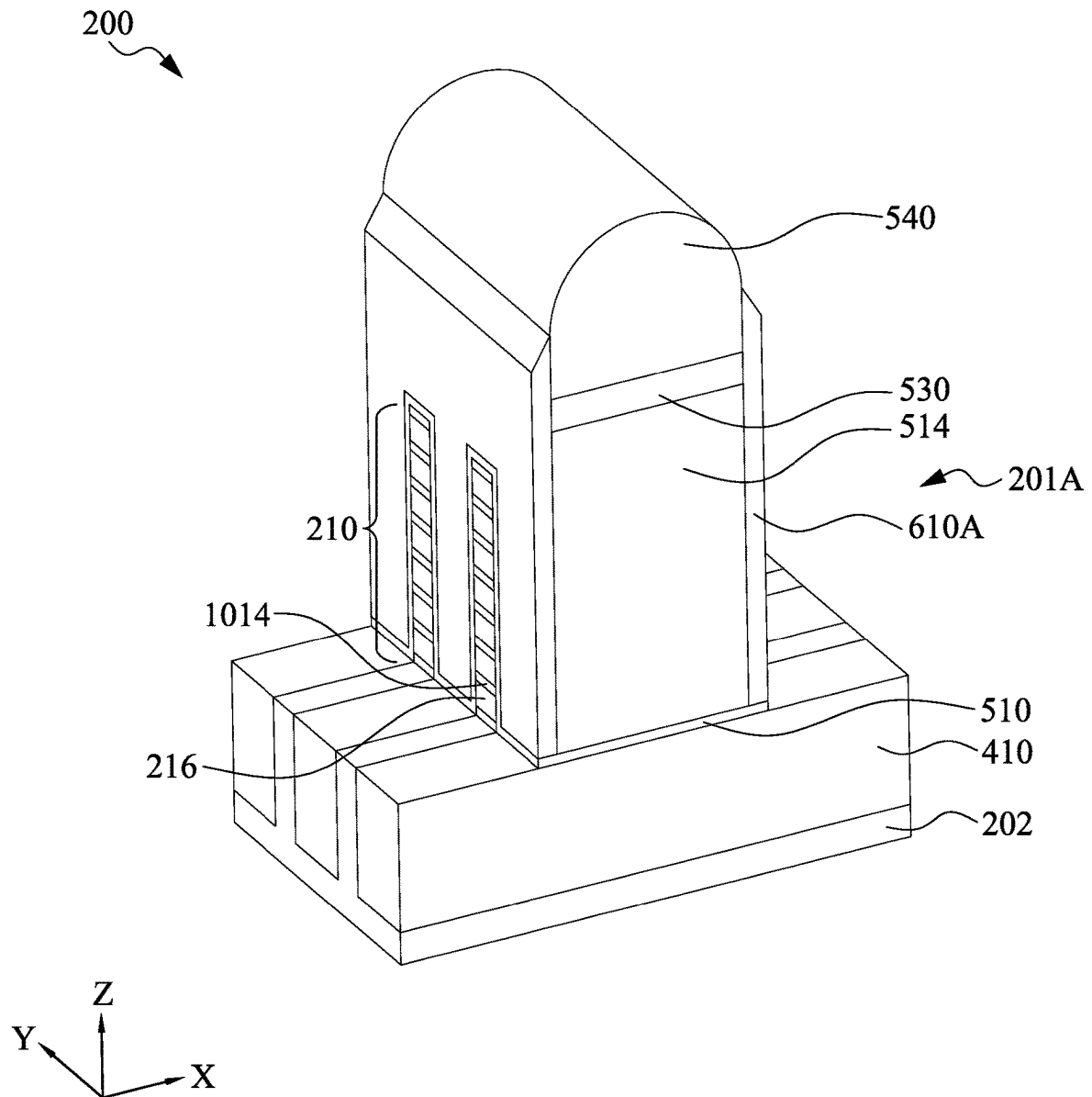


Fig. 10

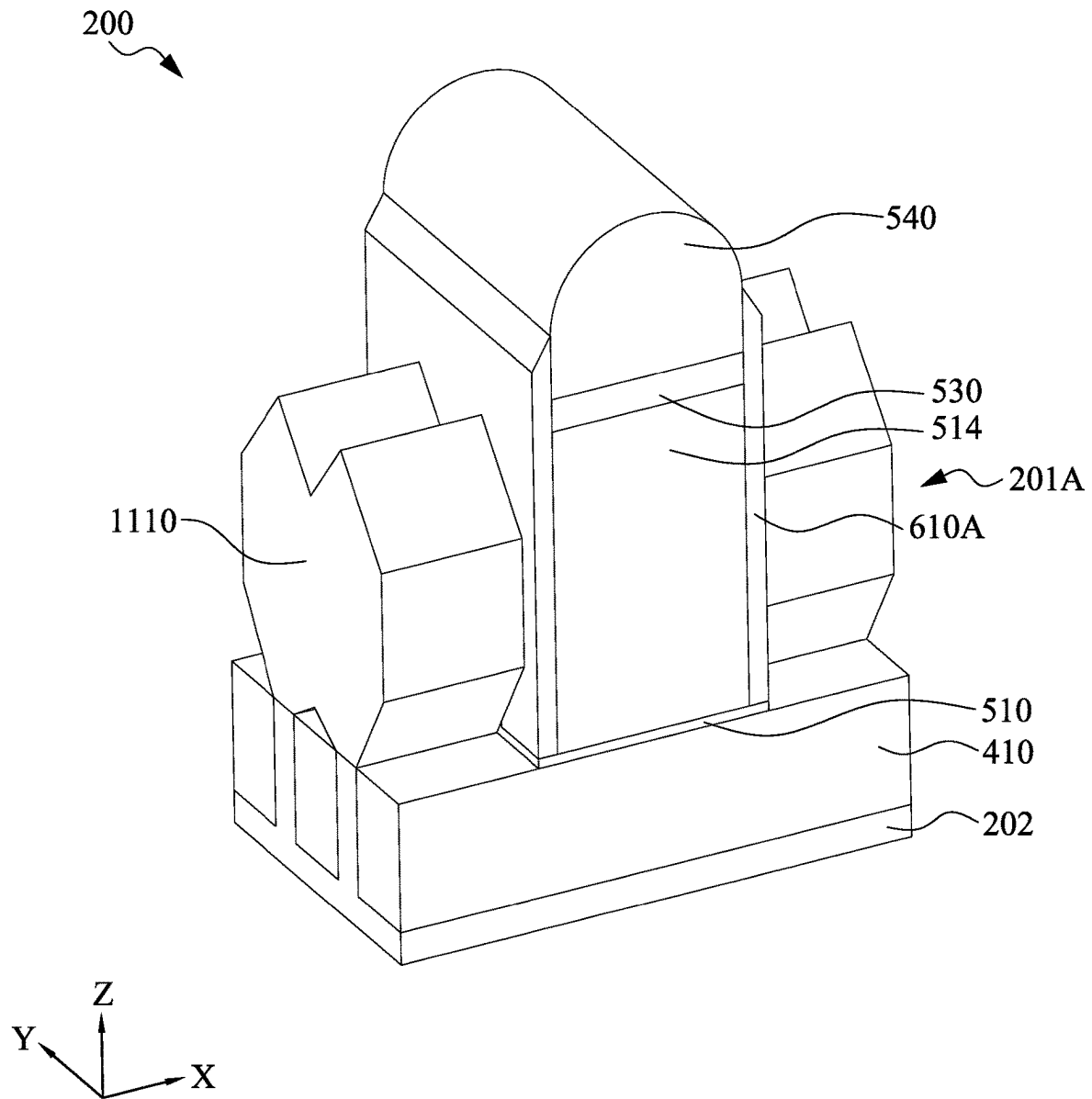


Fig. 11

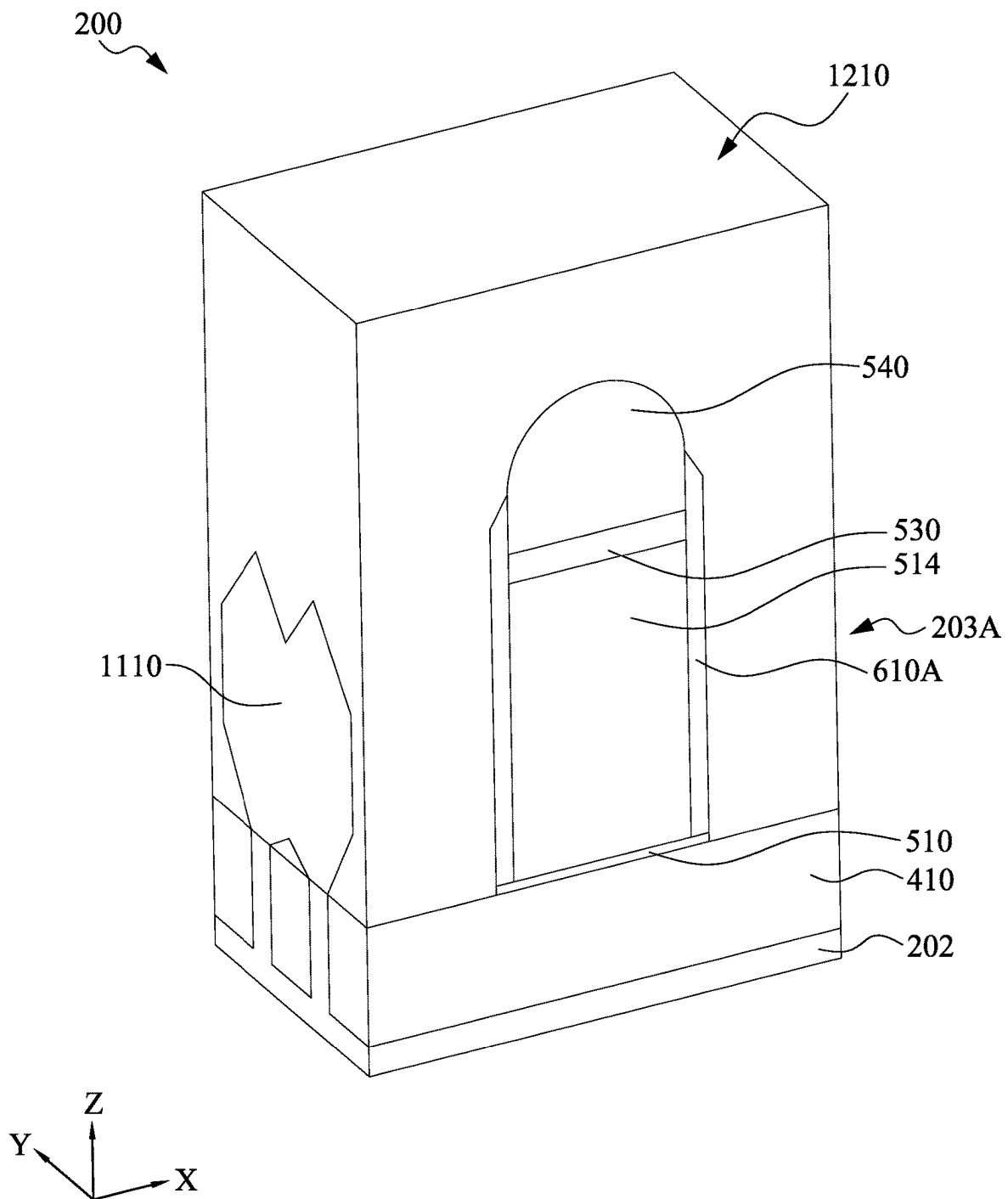


Fig. 12

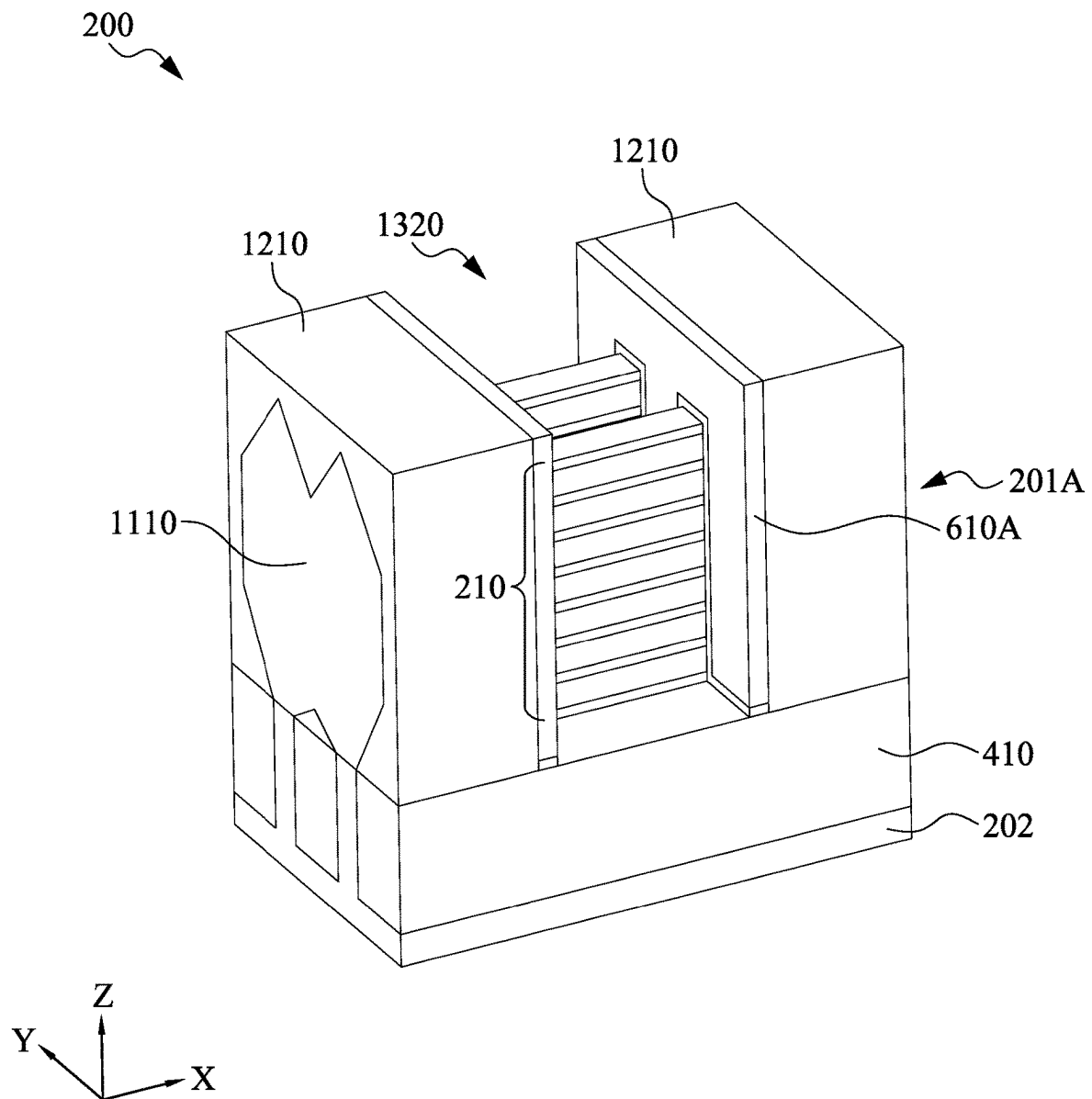


Fig. 13

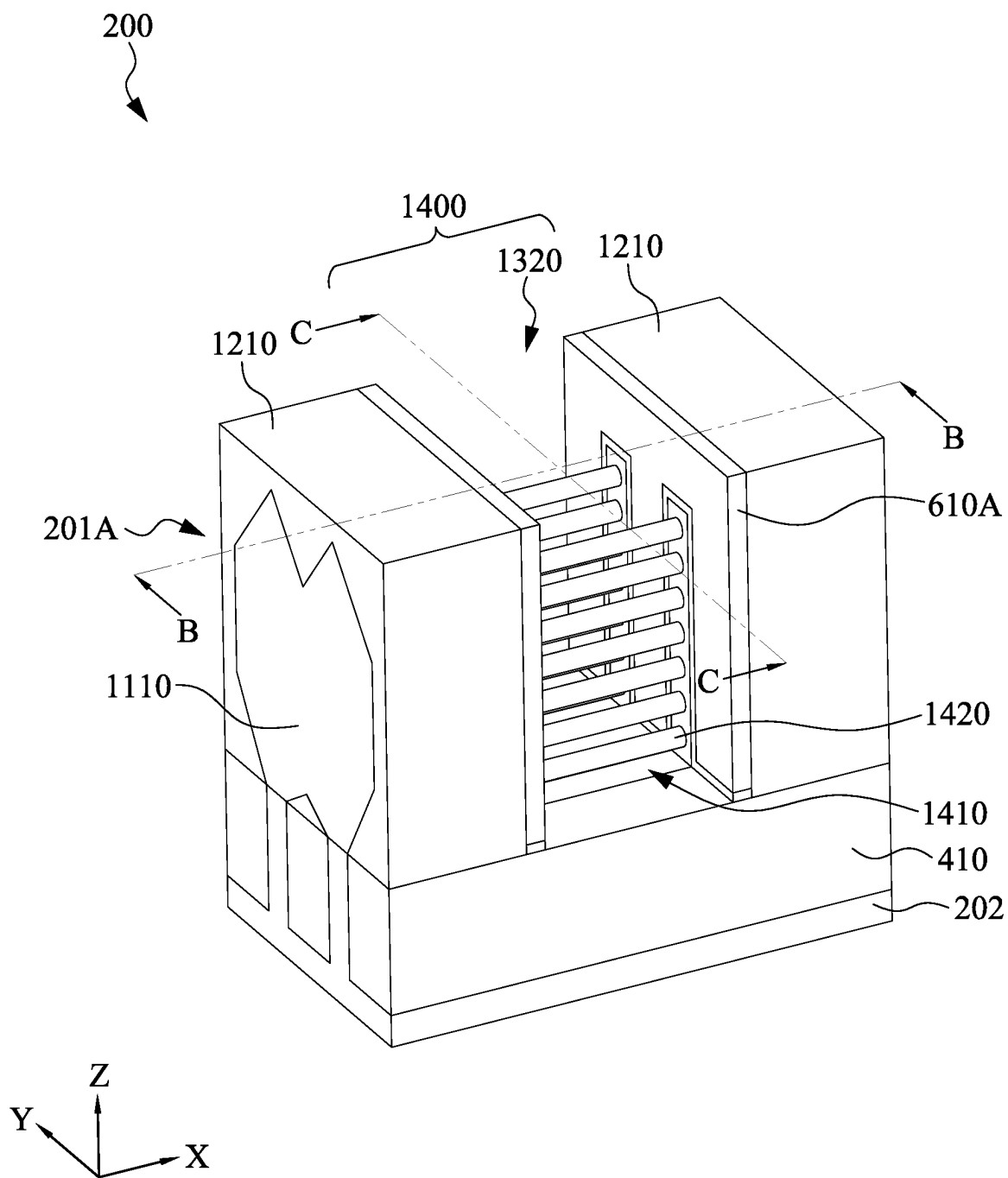


Fig. 14A

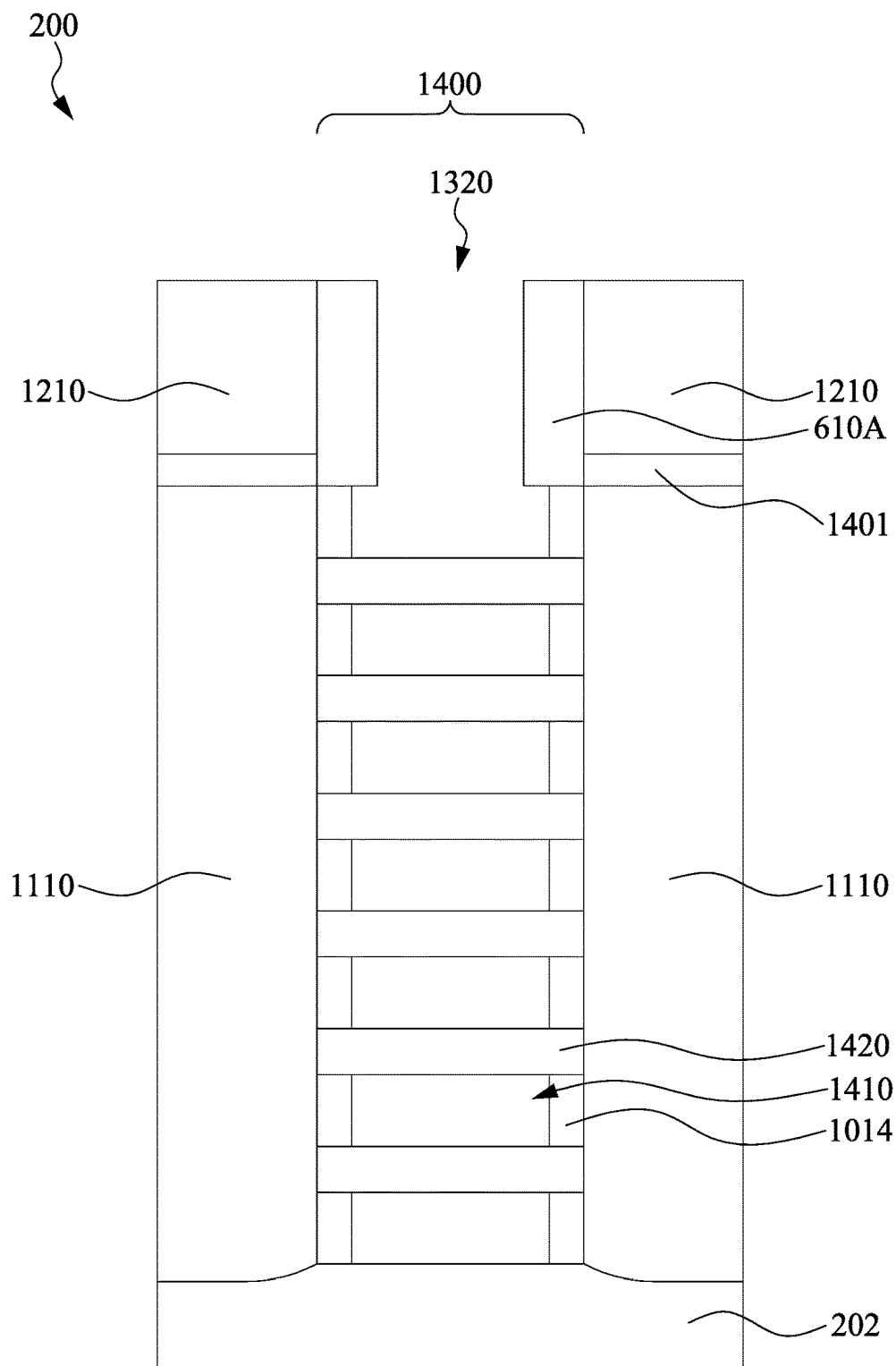


Fig. 14B

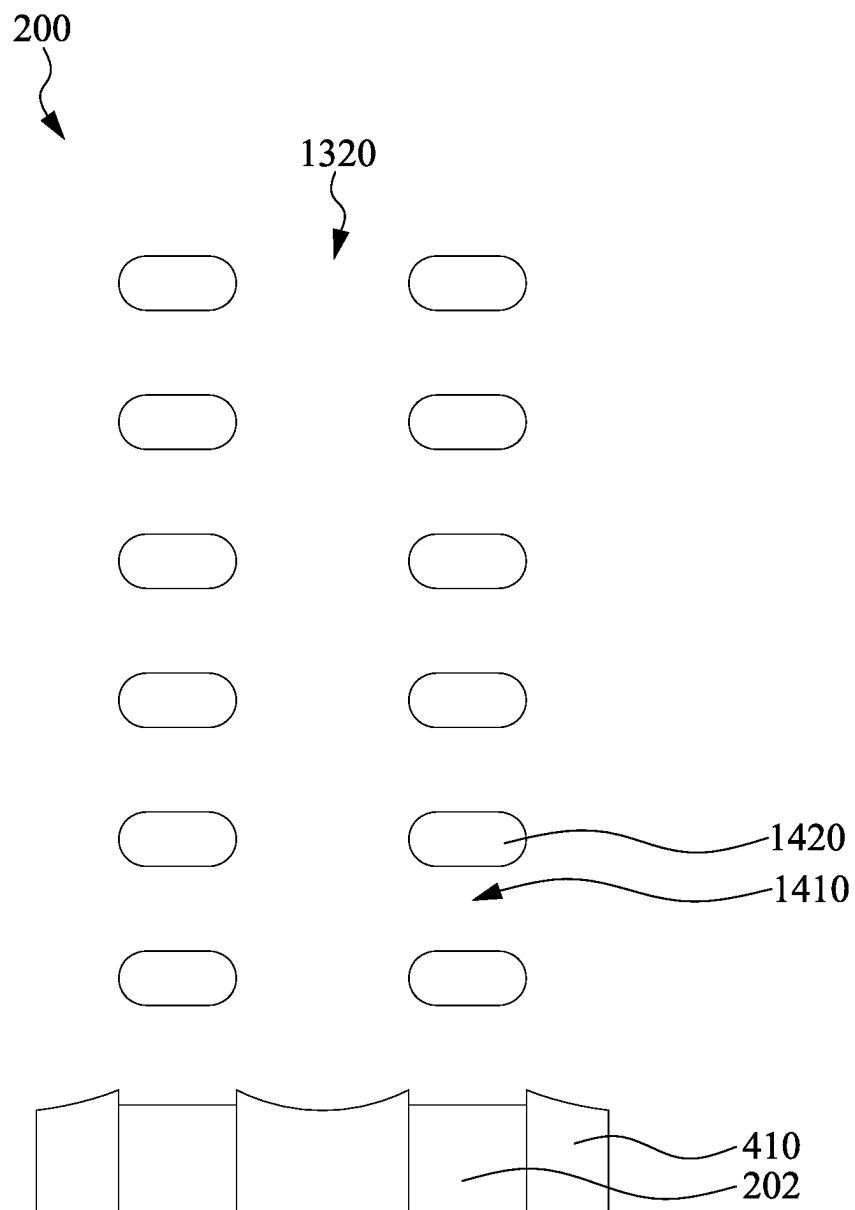


Fig. 14C

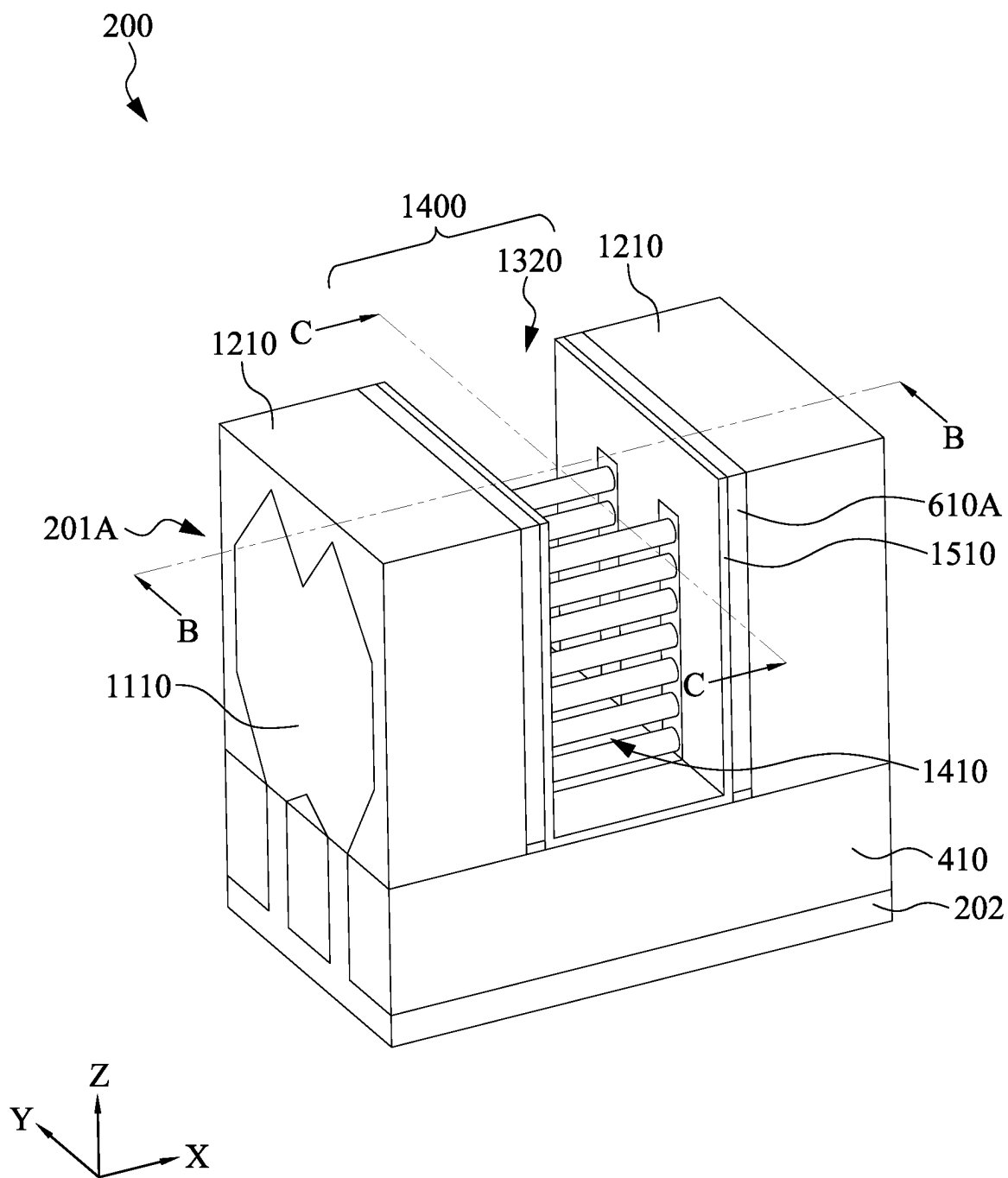


Fig. 15A

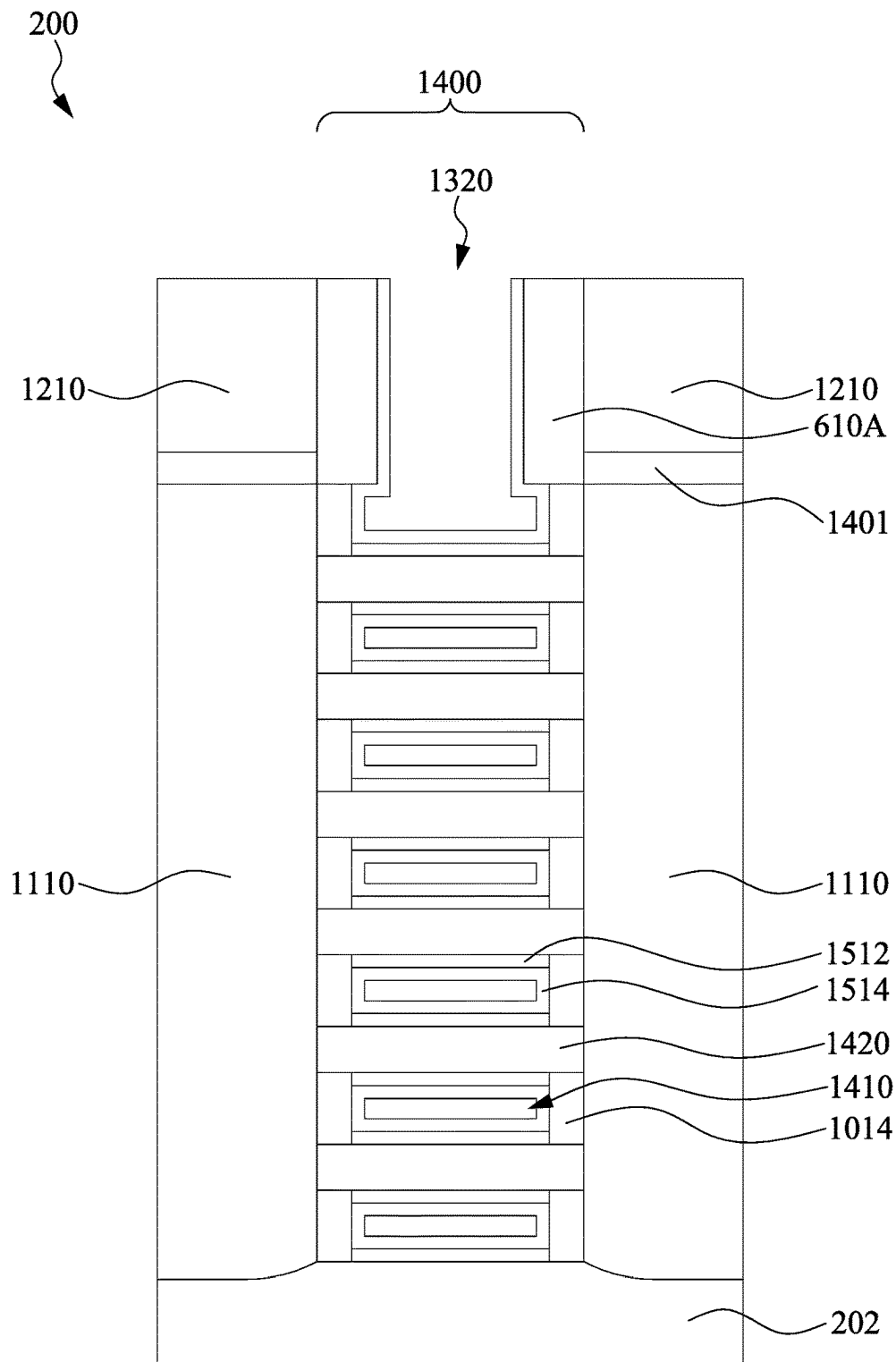


Fig. 15B

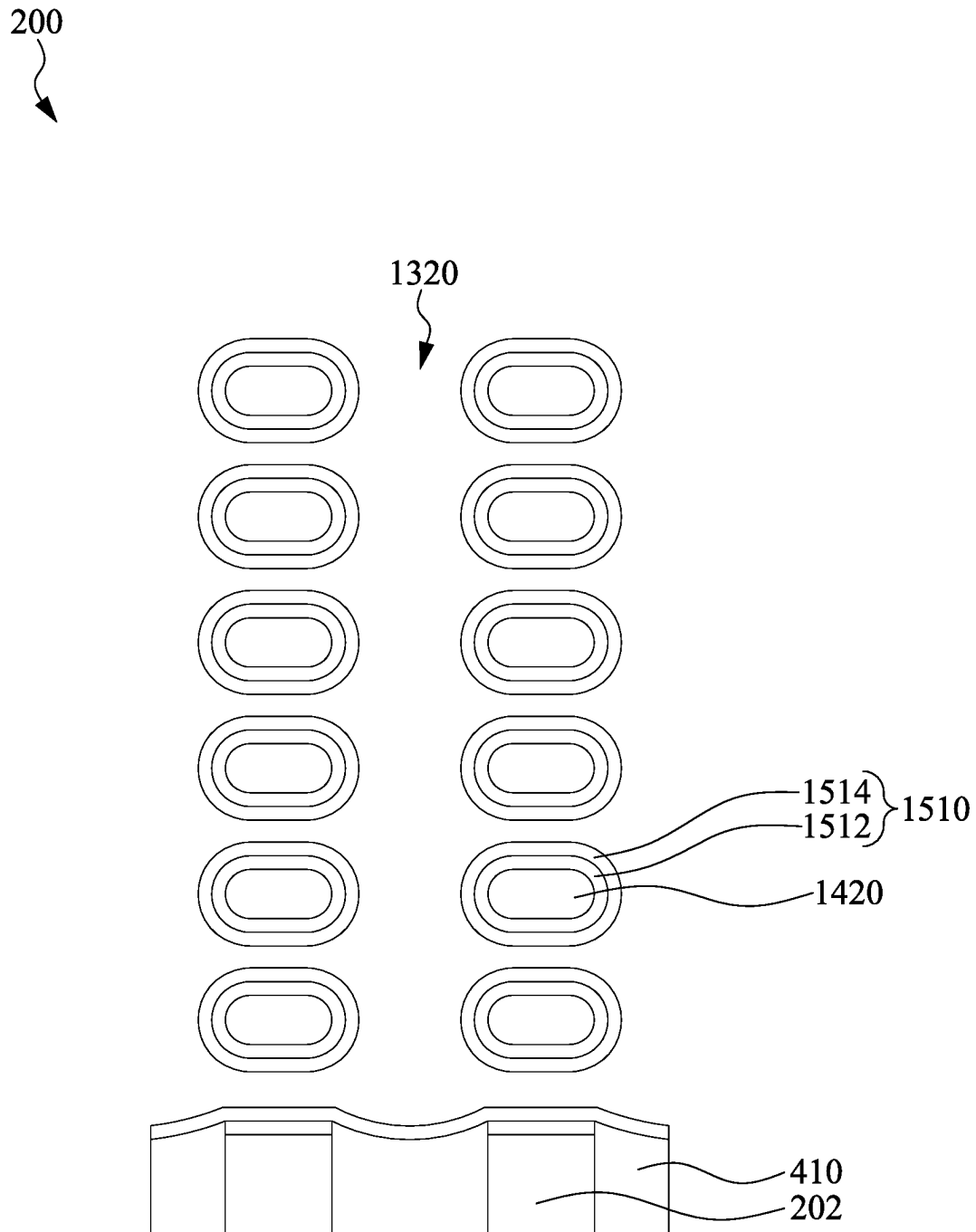


Fig. 15C

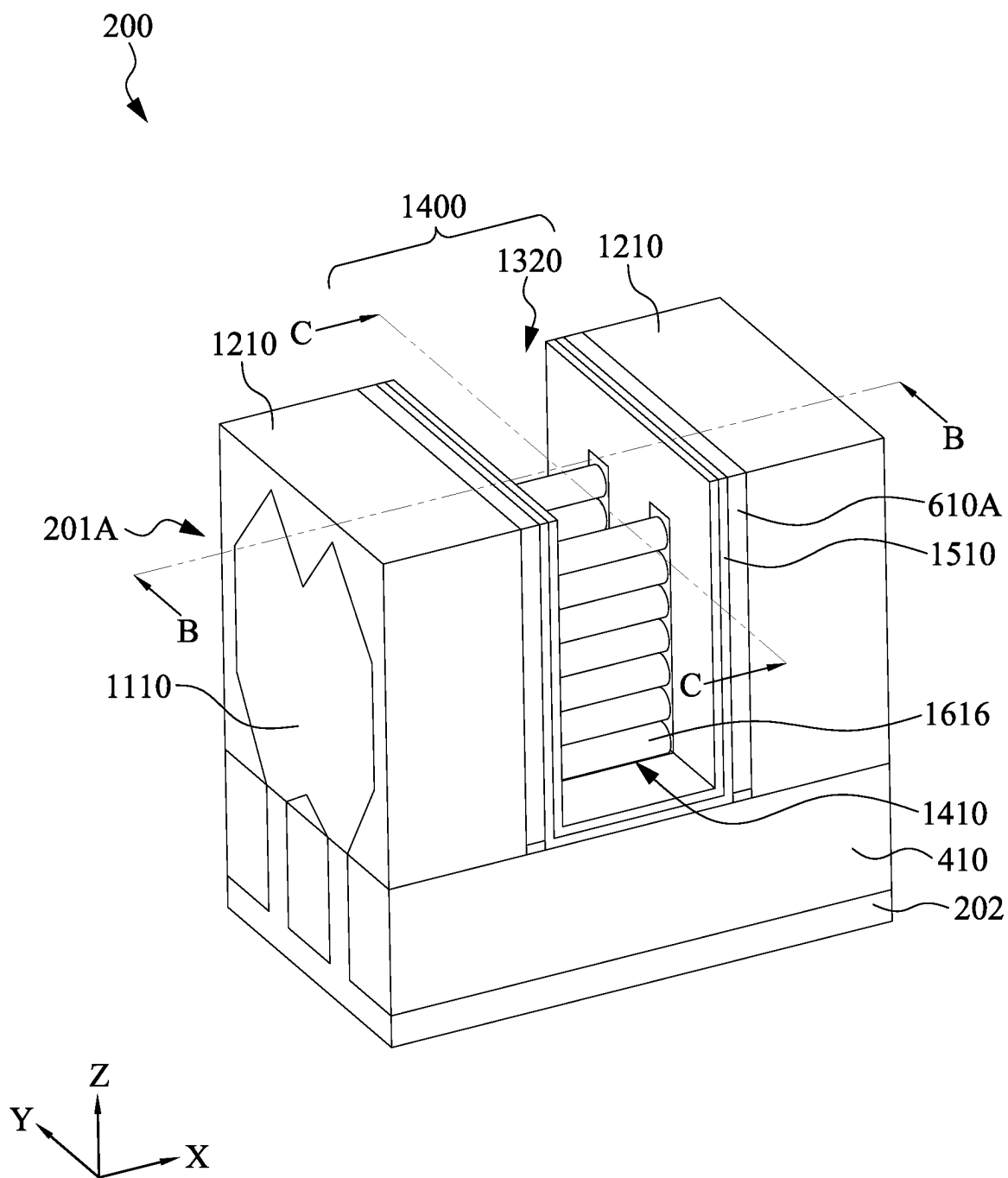


Fig. 16A

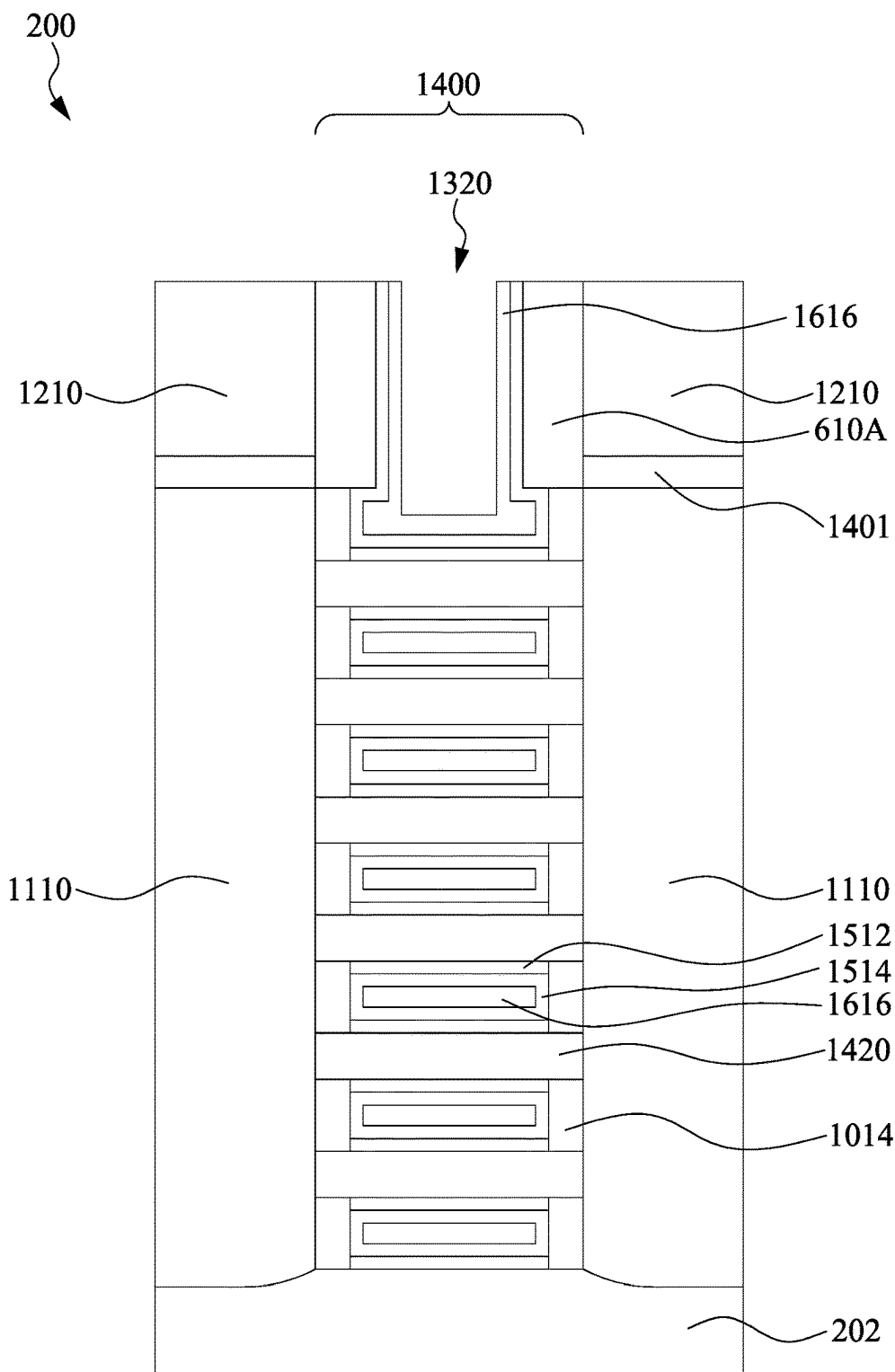


Fig. 16B

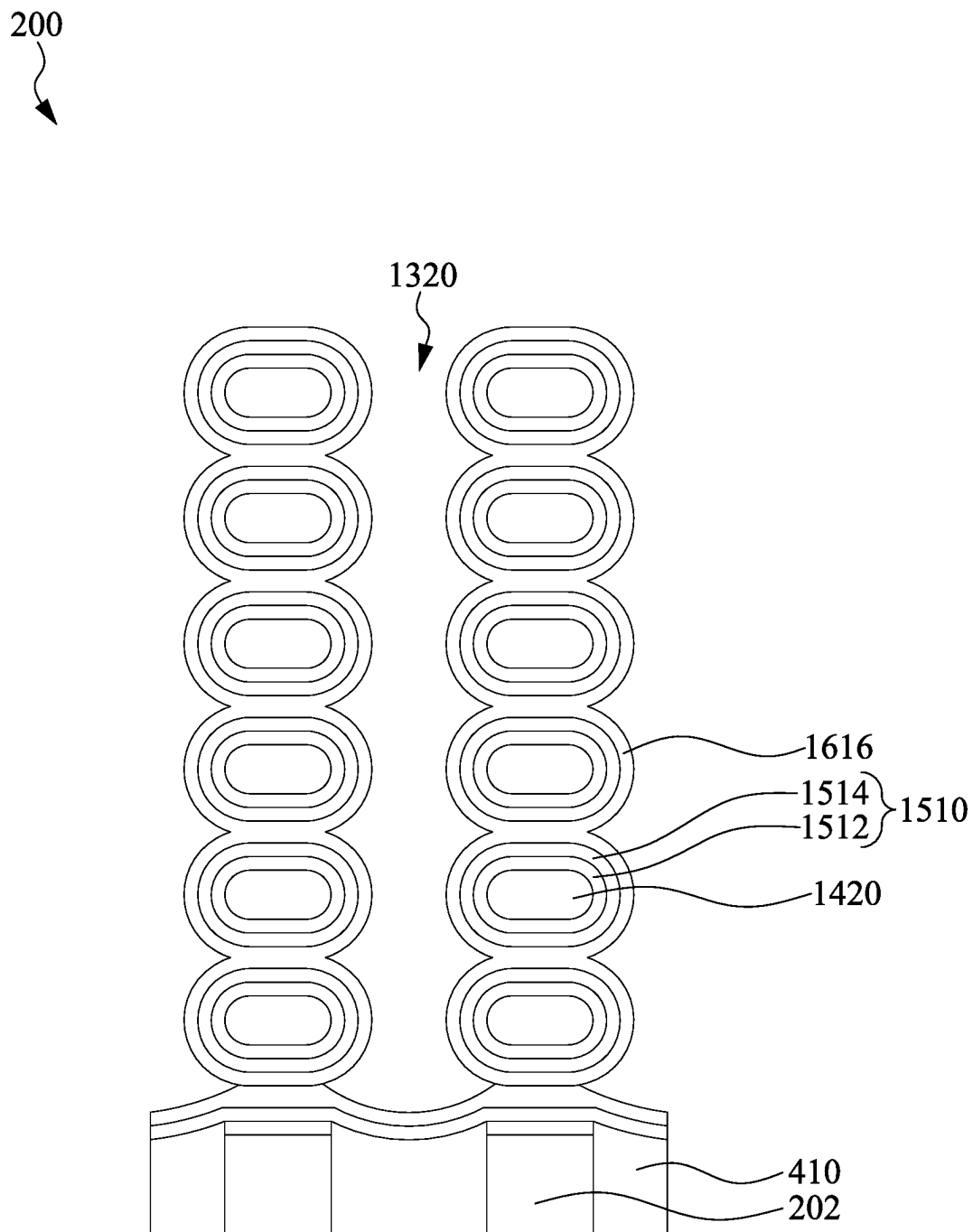


Fig. 16C

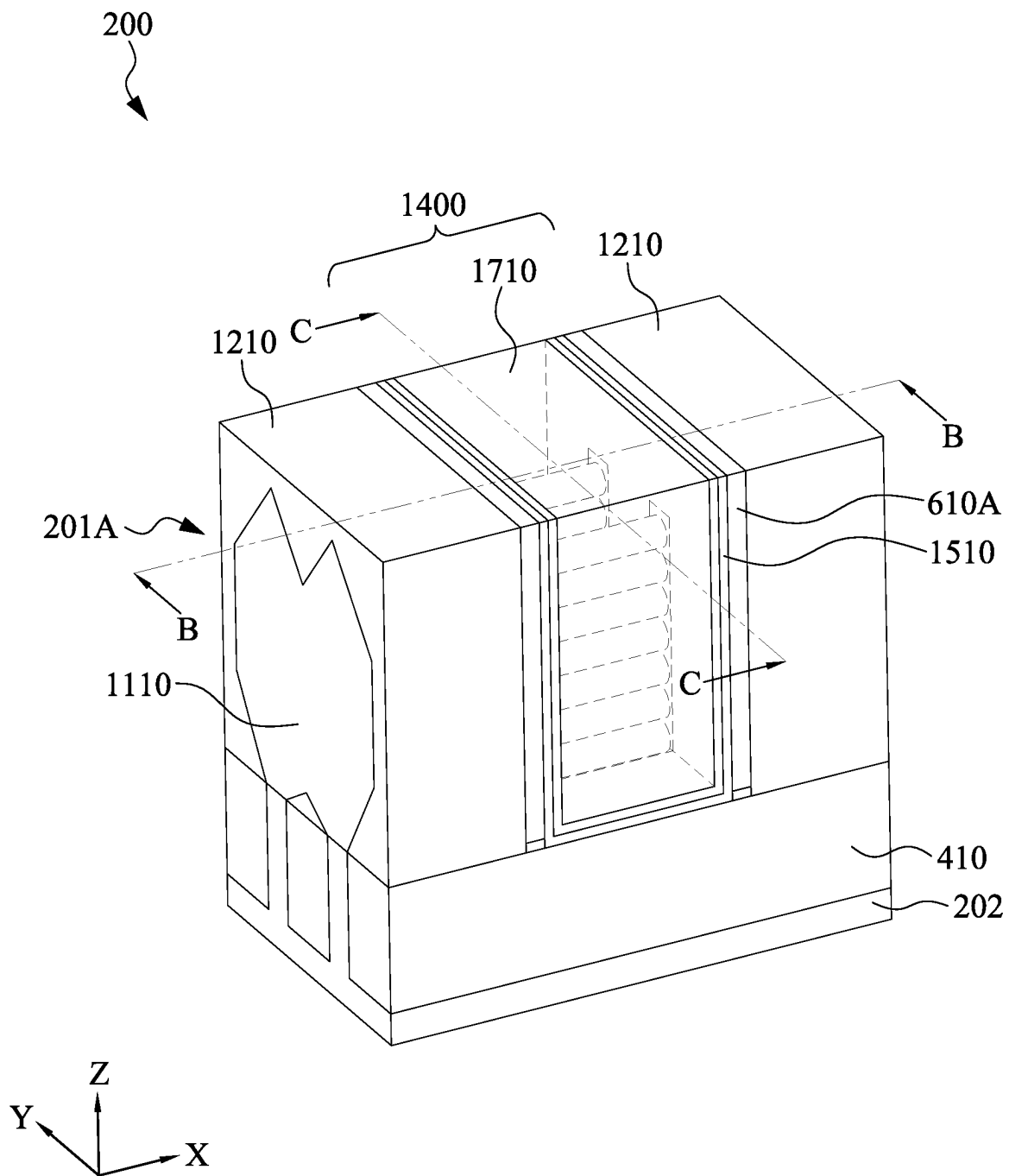


Fig. 17A

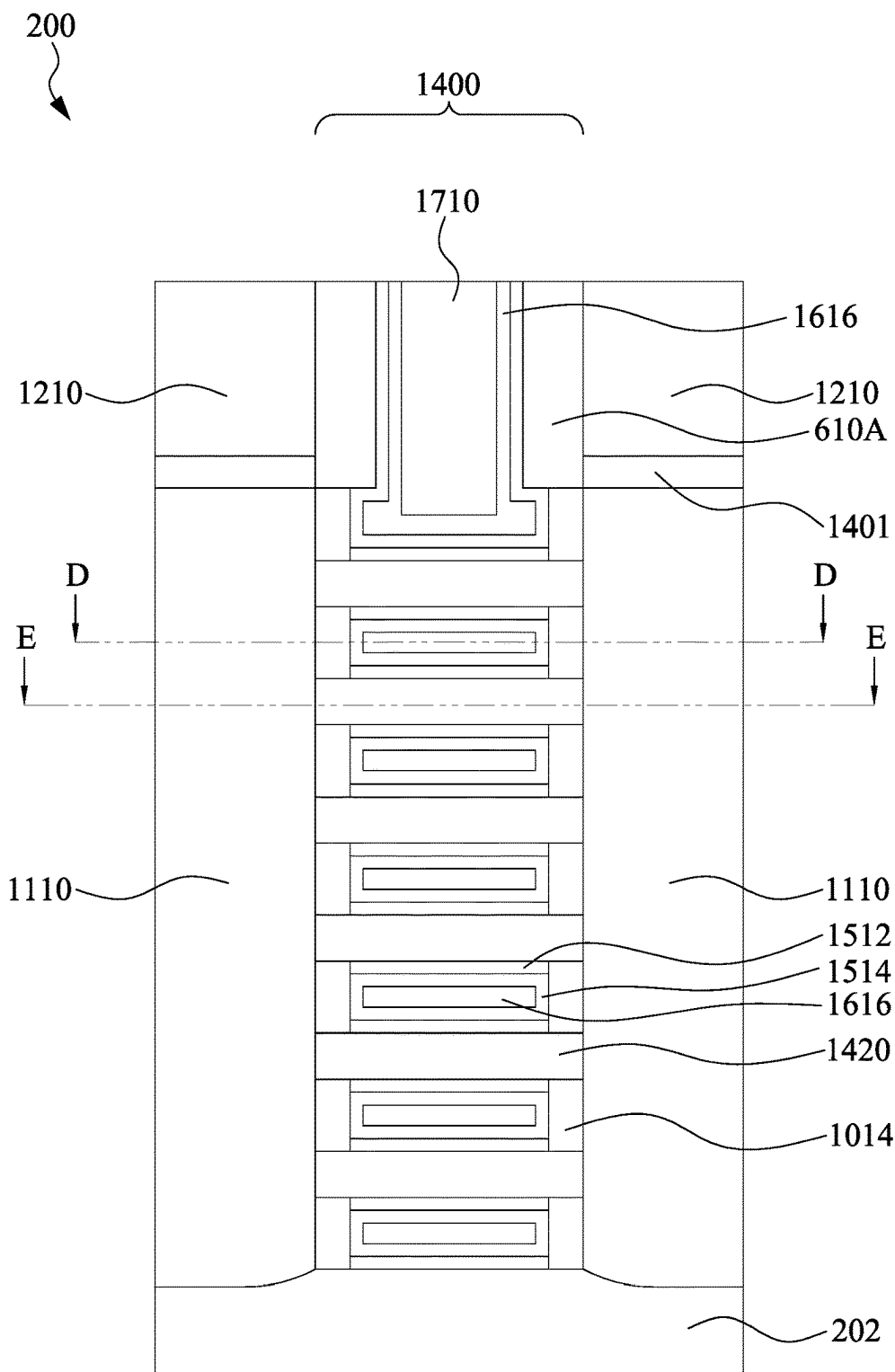


Fig. 17B

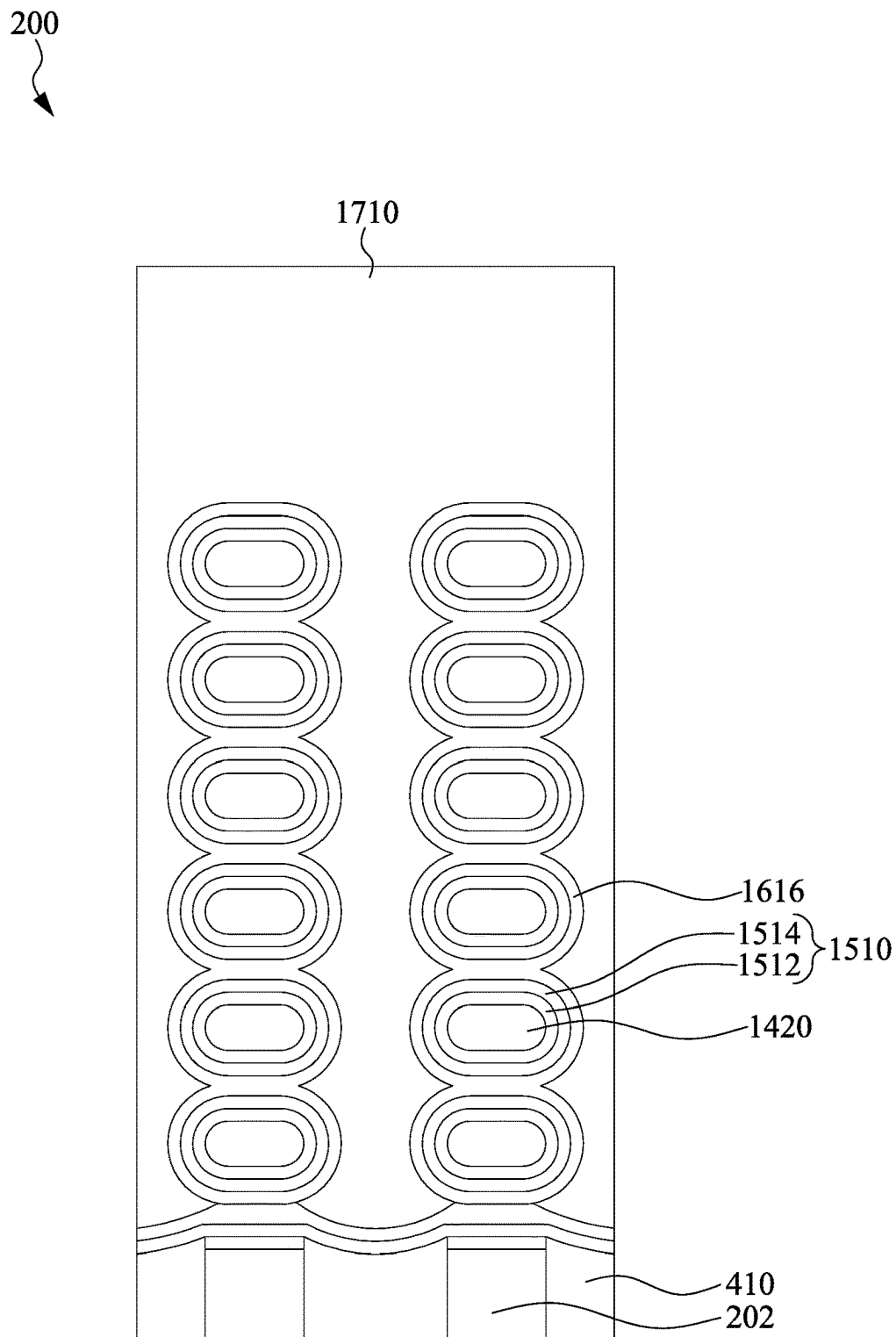


Fig. 17C

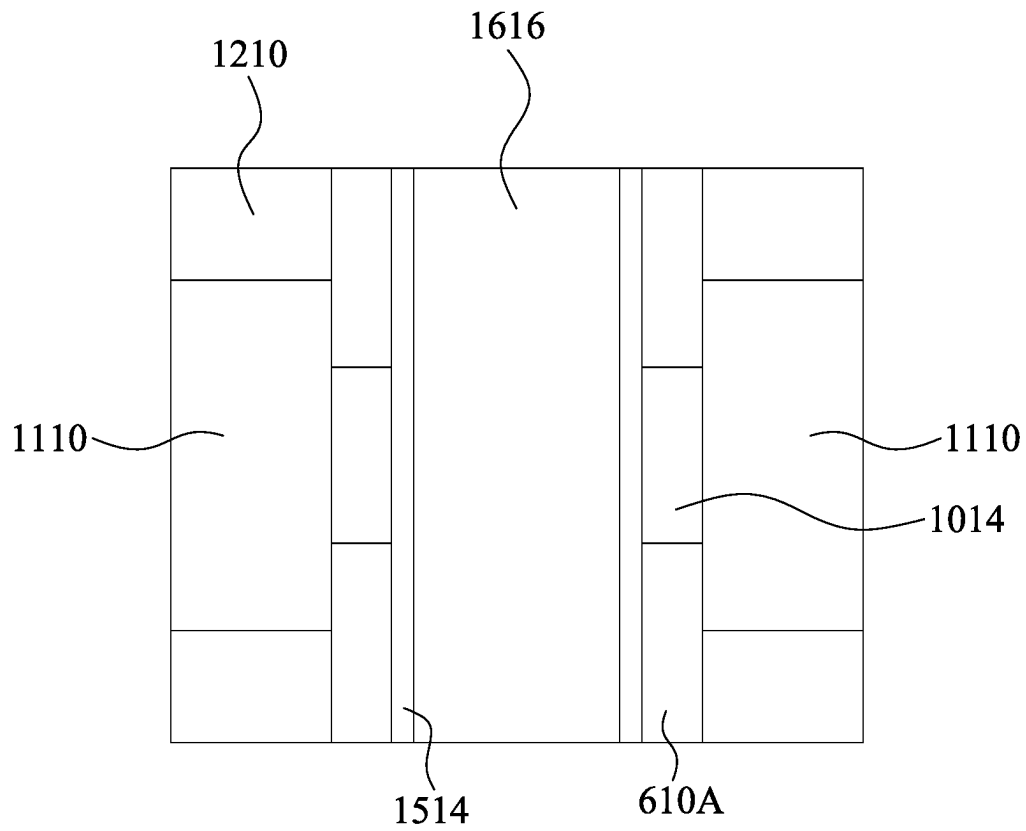


Fig. 17D

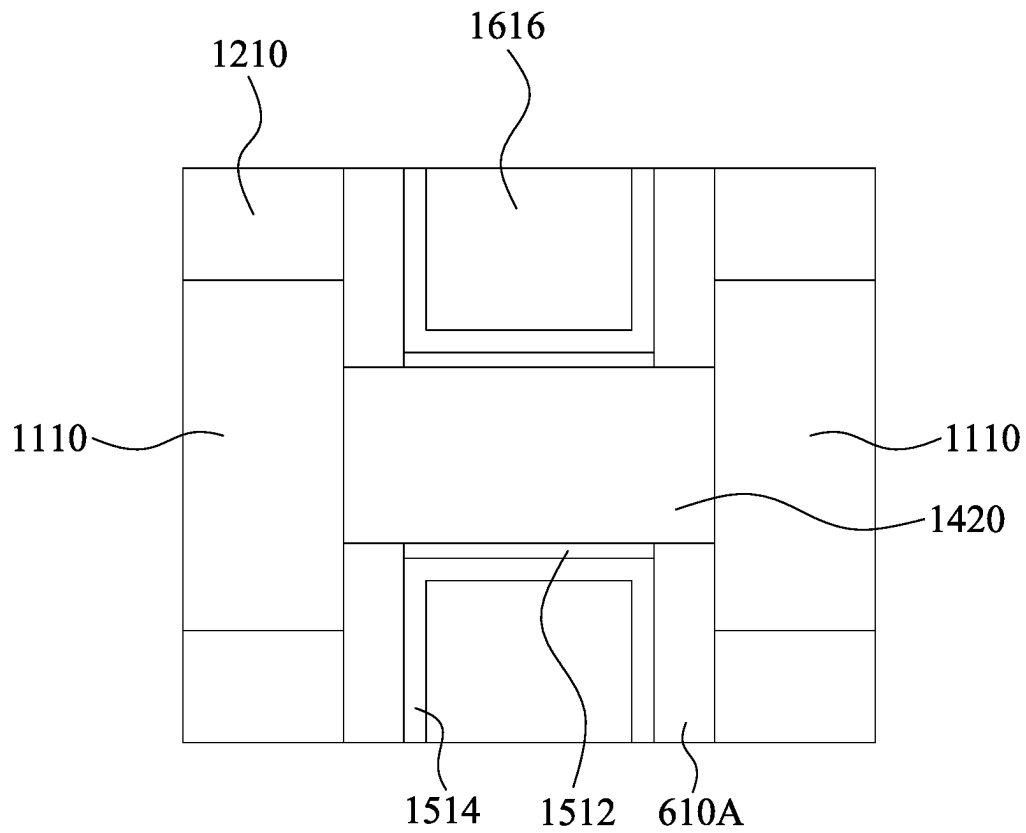


Fig. 17E

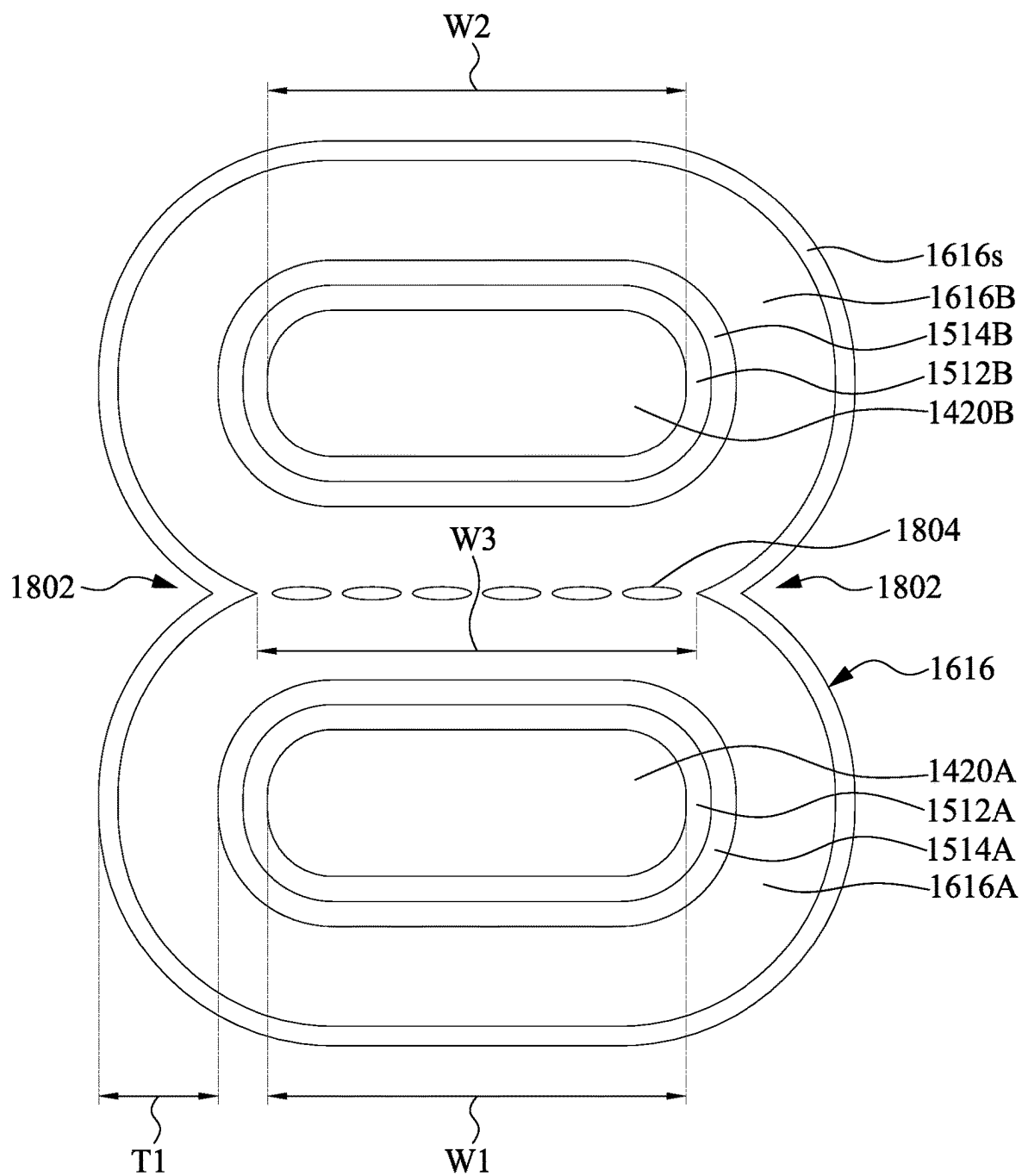


Fig. 18

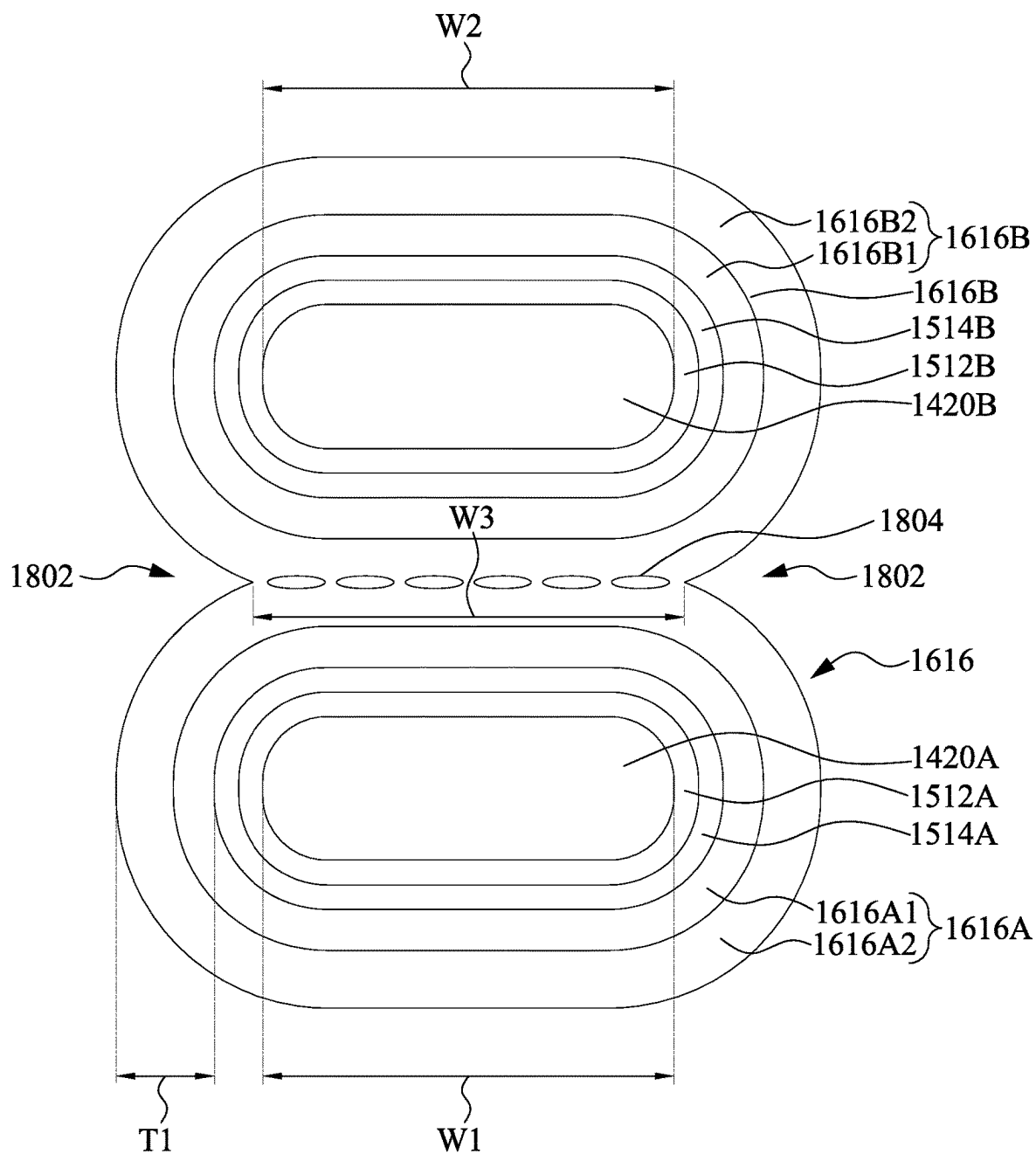


Fig. 19

SEMICONDUCTOR STRUCTURE INCLUDING NANOSHEET CHANNEL STRUCTURE AND METHOD FOR FORMING THE SAME

BACKGROUND

The electronics industry has experienced an ever increasing demand for smaller and faster electronic devices which are able to support a greater number of increasingly complex and sophisticated functions. Accordingly, there is a continuing trend in the semiconductor industry to manufacture low-cost, high-performance, and low-power integrated circuits (ICs). So far, these goals have been achieved in large part by scaling down semiconductor IC dimensions (e.g., minimum feature size) and thereby improving production efficiency and lowering associated costs. However, such scaling has also introduced increased complexity to the semiconductor manufacturing process. Thus, the realization of continued advances in semiconductor ICs and devices calls for similar advances in semiconductor manufacturing processes and technology.

Recently, gate-all around transistors (GAA) have been developed. In some examples, a gate structure of a GAA transistor is disposed around a channel region providing access to the channel on multiple sides. GAA transistors are compatible with complementary metal-oxide-semiconductor (CMOS) processes and their structure allows them to be aggressively scaled while maintaining gate control.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 shows a flow chart illustrating a method for forming a semiconductor structure in accordance with some embodiments of the present disclosure.

FIGS. 2 to 13, 14A, 15A, 16A, and 17A illustrate perspective views of a semiconductor structure at various fabrication stages in accordance with some embodiments of the present disclosure.

FIGS. 14B, 15B, 16B, and 17B illustrate cross-sectional views of a semiconductor structure taken along line B-B of FIGS. 14A, 15A, 16A, and 17A, respectively, in accordance with some embodiments of the present disclosure.

FIGS. 14C, 15C, 16C, and 17C illustrate cross-sectional views of a semiconductor structure taken along line C-C of FIGS. 14A, 15A, 16A, and 17A, respectively, in accordance with some embodiments of the present disclosure.

FIGS. 17D and 17E illustrate cross-sectional views of a semiconductor structure taken along line D-D and line E-E, respectively, of FIG. 17B in accordance with some embodiments of the present disclosure.

FIG. 18 illustrates a magnified cross-sectional view of a portion of the semiconductor structure illustrated in FIG. 17C in accordance with some embodiments of the present disclosure.

FIG. 19 illustrates a magnified cross-sectional view of a portion of the semiconductor structure illustrated in FIG. 17C in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

As used herein, the terms such as “first,” “second” and “third” describe various elements, components, regions, layers and/or sections, but these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another. The terms such as “first,” “second” and “third” when used herein do not imply a sequence, order, or importance unless clearly indicated by the context.

Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the disclosure are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in the respective testing measurements. Also, as used herein, the terms “substantially,” “approximately” or “about” generally means within a value or range (e.g., within 10%, 5%, 1%, or 0.5% of a given value or range) that can be contemplated by people having ordinary skill in the art. Alternatively, the term “substantially,” “approximately” or “about” means within an acceptable standard error of the mean when considered by one of ordinary skill in the art. People having ordinary skill in the art can understand that the acceptable standard error may vary according to different technologies. Other than in the operating/working examples, or unless otherwise expressly specified, all of the numerical ranges, amounts, values and percentages such as those for quantities of materials, durations of times, temperatures, operating conditions, ratios of amounts, and the likes thereof disclosed herein should be understood as modified in all instances by the terms “substantially,” “approximately” or “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the present disclosure and attached claims are approximations that can vary as desired. At the very least, each numerical parameter should at least be construed in light of the number of reported significant digits

and by applying ordinary rounding techniques. Ranges can be expressed herein as from one endpoint to another end point or between two endpoints. All ranges disclosed herein are inclusive of the endpoints, unless specified otherwise.

The gate all around (GAA) transistor structures may be patterned by any suitable method. For example, the structures may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used to pattern the GAA structure.

In one comparative embodiment, a semiconductor structure associated with an n-type transistor includes a work function adjustment layer (WFA layer), in which the WFA layer includes an aluminum (Al)-containing material such as TiAl, TiAlC, TiAlN, TaAl or TaAlC. The Al-containing WFA layer needs to have a relatively large thickness (e.g., more than 25 Angstrom) to achieve a desired band-edge work function (e.g., approximately 4.2 eV), which hinders the semiconductor structure from further shrinkage in size.

Some embodiments of the present disclosure therefore provide a semiconductor structure that includes a WFA layer containing silicon (Si) element and a method for forming the semiconductor structure. The WFA layer contains Si element, instead of Al element. Therefore, the WFA layer in accordance with some embodiments of the present disclosure can have a relatively small thickness to achieve the desired band-edge work function. For example, a thickness of the Si-containing WFA layer may be half of a thickness of the Al-containing WFA layer, thereby facilitating down-scaling of a semiconductor device, such as a transistor.

FIG. 1 shows a flow chart illustrating a method 10 for forming a semiconductor structure in accordance with some embodiments of the present disclosure. The method 10 includes an operation 102, in which a first channel structure and a second channel structure are formed. The second channel structure is disposed above and substantially parallel to the first channel structure. The method 10 also includes an operation 104, in which a first high-k dielectric layer is formed around the first channel structure and a second high-k dielectric layer is formed around the second channel structure. The method 10 further includes an operation 106, in which a silicon-based precursor is introduced to form a silicon-containing work function adjustment layer between the first high-k dielectric layer and the second high-k dielectric layer. The method 10 further includes an operation 108, in which a filling metal layer is formed to surround the first nanosheet channel structure, the second nanosheet channel structure, the first high-k dielectric layer, the second high-k dielectric layer and the silicon-containing work function adjustment layer. In some embodiments, the silicon-containing work function adjustment layer includes an hourglass shape around the first high-k dielectric layer and the second high-k dielectric layer. In some embodiments, the method 10 further includes oxidizing the silicon-containing work function adjustment layer prior to forming the filling metal layer.

Although this method and other methods illustrated and/or described hereinafter are illustrated as a series of acts or events, it will be appreciated that the present disclosure is

not limited to the illustrated ordering or acts. Thus, in some embodiments, the acts may be carried out in different orders than illustrated, and/or may be carried out concurrently. Further, in some embodiments, the illustrated acts or events may be subdivided into multiple acts or events, which may be carried out at separate times or concurrently with other acts or sub-acts. In some embodiments, some illustrated acts or events may be omitted, and other un-illustrated acts or events may be included.

FIGS. 2 to 13, 14A, 15A, 16A, and 17A illustrate perspective views of a semiconductor structure at various fabrication stages in accordance with some embodiments of the present disclosure. FIGS. 14B, 15B, 16B, and 17B illustrate cross-sectional views of a semiconductor structure taken along line B-B of FIGS. 14A, 15A, 16A, and 17A, respectively, in accordance with some embodiments of the present disclosure. FIGS. 14C, 15C, 16C, and 17C illustrate cross-sectional views of a semiconductor structure taken along line C-C of FIGS. 14A, 15A, 16A, and 17A, respectively, in accordance with some embodiments of the present disclosure.

Referring to FIG. 2, an epitaxial stack 210 is formed over a substrate 202 of a semiconductor structure 200. The epitaxial stack 210 includes first epitaxial layers 214 of a first composition interposed by second epitaxial layers 216 of a second composition. In some embodiments, the first composition and the second composition are different. For example, the first epitaxial layers 214 may be formed of silicon germanium (SiGe) and the second epitaxial layers 216 may be formed of silicon (Si).

The substrate 202 can be any suitable substrate, and can be processed with various features. In some embodiments, the substrate 202 is a semiconductor substrate, such as a silicon substrate. In another embodiment, the substrate 202 includes various layers, including conductive or insulating layers formed on a semiconductor substrate. In some embodiments, the substrate 202 includes various doping configurations depending on design requirements. For example, different doping profiles (e.g., n wells, p wells) may be formed on the substrate 202 in regions designed for different transistor types, such as an n-type FET (NFET), a p-type FET (PFET), and the like. The doping profiles can be formed using any suitable process, such as ion implantation of dopants and/or diffusion processes. In some embodiments, the substrate 202 may have isolation structures, such as shallow trench isolations (STIs), and the like interposing respective regions providing different transistor types. In some embodiments, the substrate 202 is formed of any suitable semiconductor material, such as germanium, silicon carbide (SiC), SiGe, diamond, a compound semiconductor, an alloy semiconductor, and the like. In some embodiments, the substrate 202 includes an epitaxial layer formed on insulator.

It is noted that seven (7) layers of the first epitaxial layers 214 and six (6) layers of the second epitaxial layers 216 are illustrated in FIG. 2 merely for illustrative purposes, and the scope of the present disclosure is not limited thereto. Any suitable number of epitaxial layers can be formed in the epitaxial stack 210. The number of epitaxy layers depends on a desired number of channel structures for respective transistors. In some embodiments, the number of second epitaxial layers 216 is between 2 and 10, for example, to form a stack of 2 to 10 channel structures.

In some embodiments, each of the first epitaxial layers 214 has a thickness range of about 2 nanometers (nm) to about 10 nm. For example, each of the first epitaxial layers 214 has a thickness range of about 4 nm to about 10 nm. In

some embodiments, each of the first epitaxial layers **214** has a relatively uniform thickness. The first epitaxial layers **214** can have same thickness or different thicknesses. In some embodiments, each of the second epitaxial layers **216** has a thickness range of about 5 nm to about 12 nm. In some embodiments, each of the second epitaxial layers **216** is substantially uniform in thickness. The second epitaxial layers **216** can have same thickness or different thicknesses. In some embodiments, the second epitaxial layers **216** are thicker than the first epitaxial layers **214**. As described in more detail below, in some examples, channel structures of a transistor are respectively formed using the second epitaxial layers **216**. In some embodiments, the thickness of the second epitaxial layers **216** is chosen based on, for example, manufacturing considerations, transistor performance considerations, and the like. In some embodiments, the thickness of the first epitaxial layers **214** are used to define a space between adjacent channel structures, and the respective thickness of the first epitaxial layers **214** is chosen based on, for example, manufacturing considerations, transistor performance considerations, and the like.

By way of example, forming the epitaxial stack **210** may be performed by a molecular beam epitaxy (MBE) process, a metal organic chemical vapor deposition (MOCVD) process, and/or other suitable epitaxial growth processes. In some embodiments, the epitaxially grown layers, such as the second epitaxial layers **216**, include the same material as the substrate **202**. In some embodiments, the first and second epitaxial layers, **214** and **216**, include a different material than the substrate **202**. As described above, in some examples, the first epitaxial layer **214** includes an epitaxially grown SiGe layer and the second epitaxial layer **216** includes epitaxially grown Si layer. In some alternative embodiments, either of the first and second epitaxial layers, **214** and **216**, include other materials such as germanium, a compound semiconductor such as SiC, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide, an alloy semiconductor such as SiGe, GaAsP, AlInAs, AlGaAs, InGaAs, GaNP, and/or GaInAsP, or combinations thereof. As described above, the materials of the first and second epitaxial layers, **214** and **216**, may be chosen based on providing differing etch selectivity properties. In various embodiments, the first and second epitaxial layers, **214** and **216**, are substantially dopant-free (i.e., having an extrinsic dopant concentration less than about $1 \times 10^7 \text{ cm}^{-3}$). For example, no intentional doping is performed during the epitaxial growth process.

As also shown in the example of FIG. 2, a hard mask (HM) layer **220** may be formed over the epitaxial stack **210**. In some embodiments, the HM layer **220** includes an oxide layer **222** (e.g., a pad oxide layer of SiO_2) and nitride layer **224** (e.g., a pad nitride layer of Si_3N_4) formed over the oxide layer **222**. In some embodiments, the oxide layer **222** includes thermally grown oxide, chemical vapor deposition (CVD) deposited oxide, and/or atomic layer deposition (ALD) deposited oxide. In some embodiments, the nitride layer **224** includes a nitride layer deposited by CVD or other suitable technique. The HM layer **220** is used to protect portions of the substrate **202** and/or the epitaxial stack **210** and/or is used to define a pattern (e.g., fin elements) as described below.

Referring to FIG. 3, a plurality of fin elements **310** (referred to as fins) are formed. The plurality of fin elements **310** protrude from the substrate **202**. In various embodiments, each of the fins **310** includes a portion formed from the substrate **202**, a portion of the epitaxial stack **210**, and a portion of the HM layer **220**. The portion of the epitaxial

stack **210** includes portions of the first epitaxial layers **214** and the second epitaxial layers **216**.

In some embodiments, the fins **310** are fabricated using suitable processes including photolithography and etch processes. During a photolithography process, in an example, a photoresist layer is formed (e.g., spun) over the surface of the semiconductor structure **200**, for example, over the HM layer **220** of FIG. 2. Subsequently, the photoresist layer is exposed according to a mask of patterns, and is developed to form the patterns in the photoresist layer. The photoresist layer with the patterns can be used as a masking element to pattern other layers. In some embodiments, patterning the photoresist layer to form the masking element is performed using an electron beam (e-beam) lithography process. The masking element is then used to protect regions of the substrate **202**, and layers formed thereupon, while an etch process forms trenches **314** in unprotected regions through the HM layer **220**, through the epitaxial stack **210**, and into the substrate **202**, thereby leaving the fins **310**. In some examples, the trenches **314** is formed using a dry etch (e.g., reactive ion etching), a wet etch, and/or combination thereof.

Numerous other embodiments of methods to form fins on a substrate can also be used. In an example, a method to form fins can include defining a fin region (e.g., by mask or isolation regions) and epitaxially growing the epitaxial stack **210** in the form of the fins **310**. In some embodiments, forming the fins **310** includes a trim process to decrease widths of the fins **310**. The trim process includes wet and/or dry etching processes.

Referring to FIG. 4, isolation structures, such as STIs, are formed between the fins **310**. By way of example, in some embodiments, a dielectric layer of dielectric materials is first deposited over the substrate **202**, filling the trenches **314** with the dielectric material. In some embodiments, the dielectric layer may include SiO_2 , silicon nitride, silicon oxynitride, fluorine doped silicate glass (FSG), a low dielectric constant (low-k dielectric) material, a combination thereof, and/or other suitable materials. In various examples, the dielectric layer may be deposited by a CVD process, a subatmospheric CVD (SACVD) process, a flowable CVD process, an ALD process, a physical vapor deposition (PVD) process, and/or other suitable process. In some embodiments, after deposition of the dielectric layer, the semiconductor structure **200** is annealed, for example, to improve quality of the dielectric layer. In some embodiments, the dielectric layer (and the subsequently formed STIs) includes a multi-layer structure, for example, having one or more liner layers.

In an example to form the STIs, after deposition of the dielectric layer, the dielectric material is thinned and planarized, for example, by a chemical mechanical polishing (CMP) process. The CMP process planarizes the top surface of the dielectric layer. In some embodiments, the CMP process also removes the HM layer **220** from each of the fins **310**. In some embodiments, removal of the HM layer **220** is performed by using a suitable etching process (e.g., a dry or a wet etching process). In some embodiment, after the planarization, the dielectric layer is recesses to form the STIs interposing the fins **310** so that the fins **310** extend above the recessed STIs **410**. In some embodiments, the recessing process includes a dry etching process, a wet etching process, and/or a combination thereof. In some embodiments, a recessing depth is controlled (e.g., by controlling an etching time) so as to result in a desired height of the exposed upper portion of the fins **310**, referred to as upper fins **310A**. In some embodiments, the height exposes each of the first and second epitaxial layers **214** and **216** of the epitaxial stack **210**.

Referring to FIG. 5, a dummy gate structure **520** is formed over the upper fins **310A**. In some embodiments, the dummy gate structure **520** is replaced by a final gate structure **1400** (see FIG. 14A) at subsequent processing steps. In some embodiments, the dummy gate structure **520** is formed over the substrate **202** and is at least partially disposed over the upper fins **310A**. Portions of the upper fins **310A** underlying the dummy gate structure **520** are referred to as channel regions.

In some embodiments, the dummy gate structure **520** includes a dummy dielectric layer **510**, an electrode layer **514**, a HM layer **530**, and a photoresist layer **540**. In some embodiments, the dummy gate structure **520** is formed by various process steps such as layer deposition, patterning, etching, as well as other suitable processing steps. Exemplary layer deposition processes include CVD (including both low-pressure CVD and plasma-enhanced CVD), PVD, ALD, thermal oxidation, e-beam evaporation, or other suitable deposition techniques, or combinations thereof. In forming the dummy gate structure **520**, for example, the patterning process includes a lithography process (e.g., photolithography or e-beam lithography) that may further include photoresist coating (e.g., spin-on coating), soft baking, mask aligning, exposure, post-exposure baking, photoresist developing, rinsing, drying (e.g., spin drying and/or hard baking), other suitable lithography techniques, and/or combinations thereof. In some embodiments, the etching process may include dry etching (e.g., reactive ion etching), wet etching, and/or other etching methods.

In some embodiments, the dummy dielectric layer **510** is formed over the upper fins **310A**. In some embodiments, the dummy dielectric layer **510** includes SiO₂, silicon nitride, a high dielectric constant (high-k) dielectric material and/or other suitable material. In some embodiments, the dummy dielectric layer **510** includes multiple layers. In various examples, the dummy dielectric layer **510** may be deposited by a CVD process, a SACVD process, a flowable CVD process, an ALD process, a PVD process, or other suitable process. By way of example, the dummy dielectric layer **510** is used to prevent damage to the upper fins **310A** by subsequent processing.

In some embodiments, the dummy dielectric layer **510** is not included in the dummy gate structure **520**. In some embodiments, the dummy dielectric layer **510** may be removed prior to the deposition of the electrode layer **514**. In some embodiments, an additional dielectric layer is included in the dummy gate structure **520**. In some examples, the additional dielectric layer includes silicon nitride, a high-k dielectric material or other suitable material. In some embodiments, the electrode layer **514** includes polycrystalline silicon (polysilicon). In some embodiments, the HM layer **530** includes an oxide layer such as a pad oxide layer of SiO₂. In some embodiments, the HM layer **530** includes the nitride layer such as a pad nitride layer that includes Si₃N₄, silicon oxynitride and/or silicon carbide. In some embodiments, the HM layer **530** includes multiple layers (e.g., an oxide layer and a nitride layer, as described above).

Referring to FIG. 6, a spacer layer **610** is formed over the substrate **202**. In some embodiments, the spacer layer **610** is a conformal dielectric layer formed over the substrate **202**. The spacer layer **610** forms spacer elements on sidewalls of the dummy gate structure **520**.

In some embodiments, the spacer layer **610** includes a dielectric material such as silicon oxide, silicon nitride, silicon carbide, silicon oxynitride, SiCN, silicon oxycarbonyl (SiOCN), and/or a combination thereof. In some

examples, the spacer layer **610** includes multiple layers, such as main spacer walls, liner layers, and the like. By way of an example, the spacer layer **610** is formed by depositing a dielectric material over the dummy gate structure **520** using processes such as, CVD process, a SACVD process, a flowable CVD process, an ALD process, a PVD process, or other suitable process. In some embodiments, the deposition is followed by an etching-back (e.g., anisotropically) of the dielectric material.

Referring to FIG. 7, portions of the upper fins **310A** that are adjacent the channel regions underlying the dummy gate structure **520** are removed using an etch process. In some embodiments, the spacer layer **610** and the dummy dielectric layer **510** are removed from exposed regions of the substrate **202** not covered by the dummy gate structure **520**. As shown in FIG. 7, a portion of the spacer layer **610** that is shown as spacer elements **610A** remains on the sidewalls of the dummy gate structure **520**. In some embodiments, the first epitaxial layers **214** includes two portions, first portions **214A** that underlie the spacer elements **610A** and are not covered by the dummy gate structure **520** and second portions **214B** that are covered by the dummy gate structure **520**. The etch process may include a dry etch (e.g., reactive ion etching), a wet etch, and/or a combination thereof.

Referring to FIG. 8, the first portions **214A** of the first epitaxial layers **214** are removed to form first gaps **810** between the second epitaxial layers **216**. Removing the first portions **214A** of the first epitaxial layers **214** can be performed by a suitable etch process, such as a wet etch, a dry etch, and/or a combination thereof. The etch process is chosen to selectively etch the first epitaxial layers **214** without substantially etching the second epitaxial layers **216**. In some examples, the second portions **214B** of the first epitaxial layers **214** remain in the channel region.

FIG. 8 illustrates the first gaps **810** in place of the removed first portions **214A** of the first epitaxial layers **214**. The first gaps **810** are filled with the ambient environment (e.g., air or N₂). In some embodiments, the first portions **214A** of the first epitaxial layers **214** are removed by a selective wet etching process. In some embodiments, the selective wet etching includes an APM etch (e.g., ammonia hydroxide-hydrogen peroxide-water mixture). In some embodiments, the selective removal includes SiGe oxidation followed by a SiGeO_x removal. For example, the oxidation is provided by O₃ clean and then SiGeO_x is removed by an etchant such as NH₄OH. In some embodiments, the first epitaxial layers **214** are SiGe and the second epitaxial layers **216** are silicon allowing for the selective removal of the first portions **214A** of the first epitaxial layers **214**.

Referring to FIG. 9, an inner spacer layer **910** is formed over the substrate **202**. In some embodiments, the inner spacer layer **910** is a conformal dielectric layer formed over the substrate **202**. In some embodiments, the inner spacer layer **910** covers the spacer elements **610A**, and fills in the first gaps **810**.

In some examples, the inner spacer layer **910** includes a dielectric material such as silicon oxide, silicon nitride, and/or a combination thereof. In some embodiments, the inner spacer layer **910** includes multiple layers. In some embodiments, the inner spacer layer **910** is formed similarly in many respects to the spacer layer **610** described above in FIG. 6.

Referring to FIG. 10, portions of the inner spacer layer **910** that are exposed outside the spacer elements **610A** are removed. The inner spacer **910** is removed from a top surface and sidewalls of the dummy gate structure **520** as well as from the substrate **202** and the STI structures **410**,

but portions of the inner spacer **910** in the spacer elements **610A** remain between the second epitaxy layers **216** as shown by remaining inner spacers **1014**. In some examples, the inner spacers **1014** fill in the first gaps **810**. In some embodiments, the inner spacer layer **910** is removed by a dry etching process, a wet etching process, and/or a combination thereof. In some embodiments, the inner spacers **1014** provide an etch stop layer during subsequent processing.

Referring to FIG. **11**, source/drain (S/D) features **1110** are formed. In some embodiments, the S/D features **1110** are configured to form S/D terminals to the transistor **201A**. In some embodiments, the S/D features **1110** are formed by performing an epitaxial growth process that provides an epitaxy material extending from the substrate **202** and covering the upper fins **310A** that underlie the dummy gate structure **520**. In some embodiments, the S/D features **1110** are formed by epitaxially growing a semiconductor material. In various embodiments, the epitaxially grown semiconductor material may include Ge, Si, GaAs, AlGaAs, SiGe, GaAsP, SiP, or other suitable material. In some embodiments, the epitaxially grown semiconductor material is in-situ doped during an epitaxial process. For example, the epitaxially grown semiconductor material is doped with boron. In some embodiments, the epitaxially grown semiconductor material is doped with carbon to form Si:C S/D features, phosphorous to form Si:P S/D features, or both carbon and phosphorous to form SiCP S/D features. In some embodiments, the second epitaxial layer **216** is silicon and the epitaxially grown semiconductor material is also silicon. In some embodiments, the second epitaxial layer **216** and the epitaxially grown semiconductor material comprise a similar material, but are differently doped. In other embodiments, the second epitaxy layer **216** includes a first semiconductor material, the epitaxially grown semiconductor material includes a second semiconductor material different than the first semiconductor material. In some embodiments, the epitaxially grown semiconductor material is not in-situ doped, and, for example, an implantation process is performed to dope the epitaxially grown semiconductor material.

Referring to FIG. **12**, an inter-layer dielectric (ILD) layer **1210** is formed over the substrate **202**. In some embodiments, the ILD layer **1210** includes materials such as tetraethylorthosilicate (TEOS) oxide, un-doped silicate glass, or doped silicon oxide such as borophosphosilicate glass (BPSG), fused silica glass (FSG), phosphosilicate glass (PSG), boron doped silicon glass (BSG), and/or other suitable dielectric materials. The ILD layer **1210** may be deposited by a PECVD process or other suitable deposition technique. In some embodiments, after formation of the ILD layer **1210**, the semiconductor structure **200** is subject to a high thermal budget process to anneal the ILD layer **1210**.

Referring to FIG. **13**, the dummy gate structure **520** is removed to expose the upper fins **310A**. In some examples, after depositing the ILD layer **1210**, a planarization process is performed to expose a top surface of the dummy gate structure **520**. For example, the planarization process includes a CMP process that removes portions of the ILD layer **1210** overlying the dummy gate structure **520** and planarizes a top surface of the semiconductor structure **200**. In addition, the CMP process removes the HM layer **530** overlying the dummy gate structure **520** to expose the electrode layer **514**. Thereafter, in some embodiments, the remaining dummy gate structure **520** is removed from the substrate **202**. In some embodiments, the electrode layer **514** and the dummy dielectric layer **510** are removed. In some alternative embodiments, the electrode layer **514** is removed

while the dummy dielectric layer **510** is not removed. The removal of the dummy gate structure **520** results in a gate trench **1320** illustrated in FIG. **13**. The dummy gate structure **520** may be removed using a selective etch process such as a selective wet etch, a selective dry etch, or a combination thereof.

In some embodiments, as shown in FIG. **13**, the dummy dielectric layer **510** is removed. The dummy dielectric layer **510** is removed similarly in many respects to the etching process described above association with FIG. **7**. In the present embodiment, the etch process is chosen to selectively etch the dummy dielectric layer **510** without substantially etching the upper fins **310A**, the spacer elements **610A**, and the STI structures **410**.

Referring to FIGS. **14A**, **14B** and **14C**, the second portions **214B** of the first epitaxial layers **214** in the gate trenches **1320** are removed to form channel structures **1420** using the remaining second epitaxial layers **216**. In some embodiments, the channel structures **1420** are nanosheet channel structures formed of a plurality of parallel nanosheets. A nanosheet in the nanosheet channel structure **1420** generally refers to a two-dimensional semiconductor slab with a length or width greater than about 100 nm and a thickness less than about 20 nm. The nanosheet may extend in a direction, e.g., the x-axis, in which the substrates **202** extends. In some embodiments, the nanosheet channel structure **1420** extends in a direction perpendicular to the direction in which the dummy gate structure **520** extends. In some embodiments, the channel structures **1420** are nanowire channel structures including a plurality of nanowire layers, in which each nanowire layer include more than one nanowire. The nanowires may include a bar shape with a width less than that of a nanosheet and a length and a thickness substantially equal to those of a nanosheet. FIG. **14B** illustrates a cross-section corresponding to FIG. **14A** along line B-B. The cross-section along line B-B is a cross-section of the semiconductor structure **200** perpendicular to a top surface of the substrate **202** and along a lengthwise direction of channel structures, and is referred to as a channel length cross-section in the present disclosure. FIG. **14C** illustrates a cross-section corresponding to FIG. **14A** along line C-C. The cross-section along line C-C is a cross-section of the semiconductor structure **200** perpendicular to a top surface of the substrate **202** and perpendicular to a lengthwise direction of channel structures, and is referred to as a channel width cross-section in the present disclosure. The second portions **214B** of the first epitaxial layers **214** are removed similarly in many respects to the etching process described above association with FIG. **8**. FIGS. **14A**, **14B** and **14C** illustrate second gaps **1410** in place of the removed second portions **214B** of the first epitaxial layers **214**. The second gaps **1410** are between the adjacent second epitaxy layers **216** in the channel region. The second gaps **1410** may be filled with the ambient environment (e.g., air or nitrogen). As a result, the second epitaxial layers **216** in the gate trench **1320** form the channel structures **1420**.

In some embodiments, the channel structure **1420** is referred to as a nanosheet or a nanowire. In some embodiments, the channel structure **1420** is bar-shaped. As shown in FIG. **14C**, the channel width cross-section of the channel structure **1420** has a rounded rectangular shape, but the disclosure of the present invention is not limited thereto. For example, the channel width cross-section of the channel structure **1420** may have an oval shape.

As described above, in some examples, the inner spacers **1014** serve as etch stop layers to protect the S/D features

11

1110 during removal of the second portions 214B of the first epitaxial layers 214 in the gate trench 1320.

Referring to FIG. 14B, in some embodiments, an additional dielectric layer 1401 is formed above the S/D feature 1110. In some examples, the additional dielectric layer 1401 is formed similarly in many respects to the process described above association with FIG. 6, thus a detailed description is omitted for clarity purposes.

In some embodiments, a final gate structure 1400, also referred to as a metal-gate structure or replacement gate structure, is subsequently formed over the channel structures 1420. Channels are formed in the respective channel structures 1420 when a suitable voltage is applied to the final gate structure 1400. Thus, the final gate structure 1400 forms a gate associated with the channels. In some embodiments, the final gate structure 1400 includes a gate dielectric layer and a metal gate that includes a plurality of metal or non-metal layers.

Referring to FIGS. 15A, 15B and 15C, a gate dielectric layer 1510 of the final gate structure 1400 is formed. FIG. 15B illustrates the channel length cross-section corresponding to FIG. 15A along line B-B. FIG. 15C illustrates the channel width cross-section corresponding to FIG. 15A along line C-C. In some embodiments, the gate dielectric layer 1510 includes multiple layers. In some examples, the gate dielectric layer 1510 includes an interfacial layer 1512 and a high-k dielectric layer 1514. In some embodiments, the interfacial layer 1512 is disposed around each channel structure 1420, and the high-k dielectric layer 1514 is disposed around the respective interfacial layer 1512, as shown in FIGS. 15B and 15C. After the formation of the gate dielectric layer, the second gaps 1410 between the channel structures 1420 are reduced.

In some embodiments, the interfacial layer 1512 includes a dielectric material such as silicon dioxide (SiO_2), hafnium silicon oxide (HfSiO), or silicon oxynitride (SiON). The interfacial layer 1512 is formed by chemical oxidation, thermal oxidation, ALD, CVD, and/or other suitable method. The high-k gate dielectric layer 1514 may include suitable high-k dielectrics, such as hafnium oxide, aluminum oxide, titanium oxide, hafnium zirconium oxide, tantalum oxide, hafnium silicon oxide, zirconium oxide, zirconium silicon oxide, lanthanide oxide, or any other suitable material, or a combination thereof. In some embodiments, the high-k gate dielectric layer 1514 includes hafnium oxide (HfO_2). In some embodiments, the high-k gate dielectric layer 1514 includes HfO_2 formed from precursors, e.g., $\text{HfCl}_4 + \text{H}_2\text{O}$. In some embodiments, the high-k gate dielectric layer 1514 is formed by ALD, PVD, CVD, oxidation, and/or other suitable methods.

Referring to FIGS. 16A, 16B and 16C, a work function adjustment (WFA) layer 1616 of the metal gate is formed. FIG. 16B illustrates the channel length cross-section corresponding to FIG. 16A along line B-B. FIG. 16C illustrates the channel width cross-section corresponding to FIG. 16A along line C-C. The WFA layer 1616 is formed in the final gate structure 1400 of an NFET. The WFA layer 1616 may be formed to cover the gate dielectric layer 1510. In some embodiments, the WFA layer 1616 fills the space of the gap 1410 between adjacent channel structures 1420 and therefore forms a contiguous material of the WFA layer 1616 across different nanosheets or nanowires of the individual channel structures 1420.

In some embodiments, the WFA layer 1616 is a single layer. In some embodiments, the WFA layer 1616 contains non-metal elements, such as a semiconductor element. In some embodiments, the WFA layer 1616 includes silicon

12

element. For example, the WFA layer 1616 may include a silicon-containing material formed of only silicon. In some embodiments, the WFA layer 1616 includes transition metal silicide, but the scope of the present disclosure is not limited thereto. Examples of the transition metal silicide include, but are not limited to, titanium silicide, tantalum silicide, tungsten silicide, ruthenium silicide, or a combination thereof. In some embodiment, the WFA layer 1616 comprises a combination of silicon and transition metal silicide. In some embodiments, the WFA layer 1616 in a single layer of pure silicon. The pure silicon refers to a material formed of silicon without other intentionally added chemical elements during the formation of the pure silicon material. In some examples, the pure silicon contains only silicon. In some embodiments, the WFA layer 1616 is substantially free of non-silicon element. In some embodiments, the pure silicon of the WFA layer 1616 is amorphous. In some embodiments, the WFA layer 1616 is formed of only amorphous silicon. In some embodiments, a surface of the WFA layer 1616 may be oxidized, for example, by exposure to air, and thus the WFA layer 1616 may include a pure silicon covered or wrapped around by a thin film of silicon oxide. In some embodiments, the WFA layer 1616 is substantially free of metallic components, such as aluminum. In some embodiments, the WFA layer 1616 fills in the second gaps 1410 between the adjacent channel structures 1420, as shown in FIGS. 16B and 16C. In some embodiments, the WFA layer 1616 is disposed around the high-k dielectric layer 1514.

The WFA layer 1616 in accordance with the present disclosure contains silicon element. For example, the WFA layer 1616 includes pure silicon or transition metal silicide. Therefore, the semiconductor structure 200 with the WFA layer 1616 can achieve a desired band-edge work function (e.g., about 4.2 eV) with a relatively small thickness of the WFA layer 1616, e.g., less than about 15 Angstrom. With introduction of the silicon element to the WFA layer 1616, the dimension of the WFA layer 1616 can be further reduced. In some embodiments, a thickness of the WFA layer 1616 is no more than about 15 Angstrom. Therefore, the WFA layer 1616 is suitable for use in advanced technology nodes.

In some embodiments, the WFA layer 1616 may be formed by CVD or other suitable process. In some embodiments, the WFA layer 1616 is formed from a silicon-containing precursor. Examples of the silicon-containing precursor include, but are not limited to, silane (SiH_4), disilane (Si_2H_6), or the like. In some embodiments, the WFA layer 1616 is formed by a CVD operation at a temperature in a range of between approximately 300° C. and approximately 600° C. for a soak time of between approximately 10 seconds and approximately 30 minutes. In some embodiments, the WFA layer 1616 is formed from a silicon-containing precursor and a transition metal precursor. Examples of the transition metal precursor include, but are not limited to, transition metal chloride, transition metal fluoride, or the like, such as titanium chloride, tantalum chloride or tungsten fluoride. In some embodiments, after the WFA layer 1616 is formed, the semiconductor structure 200 may be annealed, for example, to improve the quality of the WFA layer 1616. In some embodiments, the WFA layer 1616 may be exposed to air and a surface of the WFA layer 1616 may be oxidized.

Referring to FIGS. 17A, 17B, 17C, 17D and 17E, a filling metal layer 1710 of the metal gate is formed. FIG. 17B illustrates the channel length cross-section corresponding to FIG. 17A along line B-B. FIG. 17C illustrates the channel width cross-section corresponding to FIG. 17A along line C-C. FIGS. 17D and 17E illustrate cross-sectional views of

13

the semiconductor structure **200** taken along line D-D and line E-E, respectively, of FIG. 17B. The cross section of FIG. 17D shows a top view of a plane in the final gate structure **1400** through the WFA layer **1616** between adjacent channel structures **1420**, while The cross section of FIG. 17E is a top view of a plane in the final gate structure **1400** through the channel structure **1420**. The filling metal layer **1710** is disposed on the WFA layer **1616**. In some embodiments, the filling metal layer **1710** covers the WFA layer **1616**, as shown in FIGS. 17A and 17C. In some embodiments, due to the oxidation of the WFA layer **1616** as described above, an oxide layer, such as a silicon oxide layer, is at an interface between the WFA layer **1616** and the filling metal layer **1710**. In some embodiments, after the forming of the filling metal layer **1710**, the filling metal layer **1710** fills the gate trench **1320**. In various embodiments, the filling metal layer **1710** may be formed by ALD, PVD, CVD, e-beam evaporation, or other suitable process. In some embodiments, the filling metal layer **1710** may include a multi-layer structure, such as various combinations of a liner layer, a wetting layer, an adhesion layer, a barrier layer, a block layer, a capping layer, or the like. By way of example, the filling metal layer **1710** may include Ti, Ag, Al, TiAl, TiAlN, Ta, TaN, TaC, TaCN, TaSiN, Mn, Zr, TiN, TaN, Ru, Mo, WN, Cu, W, Re, Ir, Co, Ni, other suitable metal materials or a combination thereof. In some embodiments, the filling metal layer **1710** includes a tungsten-based material formed from precursors, for example, $WF_6 + H_2$ or $WF_6 + B_2H_6$ or $WF_6 + SiH_4$ or $WF_6 + Si_2H_6$. In some embodiments, the filling metal layer **1710** includes a TiN layer formed from precursors, e.g., $TiCl_4 + NH_3$.

Referring to FIG. 17E, the channel structure **1420** is formed between the opposite S/D features **1110**. Referring to FIG. 17D, as described above, in some examples, the inner spacers **1014** fill the spaces in the gate trench **1320** between the adjacent channel structures **1420** and serve as etch stop layers to protect the S/D features **1110** during removal of the second portions **214B** of the first epitaxial layers **214** in the gate trench **1320**. As a result, during the forming of the final gate structure **1400**, the gate dielectric layer **1510** covers the inner spacers **1014**.

FIG. 18 illustrates a magnified cross-sectional view of a portion of the semiconductor structure **200** illustrated in FIG. 17C in accordance with some embodiments of the present disclosure. The portion of the semiconductor structure **200** includes a first channel structure **1420A**, a first high-k dielectric layer **1514A** surrounding the first channel structure **1420A**. The portion of the semiconductor structure **200** also includes a second channel structure **1420B** substantially parallel to the first channel structure **1420A**, a second high-k dielectric layer **1514B** surrounding the second channel structure **1420B**. In some embodiments, the first channel structure **1420A** is disposed above the semiconductor substrate **202** as described above, and the second channel structure **1420A** is disposed above and substantially parallel to the first channel structure **1420A**. The portion of the semiconductor structure **200** further includes a WFA layer **1616** on the first high-k dielectric layer **1514A** and the second high-k dielectric layer **1514B**. The WFA layer **1616** contains silicon element. A portion of the WFA layer **1616** is disposed between the first high-k dielectric layer **1514A** and the second high-k dielectric layer **1514B**. In some embodiments, the WFA layer **1616** is in physical contact with the first high-k dielectric layer **1514A** and the second high-k dielectric layer **1514B**. In some embodiments, the WFA layer **1616** is in direct contact with the first high-k dielectric layer **1514A** and the second high-k dielectric layer

14

1514B. In some embodiments, the WFA layer **1616** includes a first portion **1616A** around the first high-k dielectric layer **1514A** and a second portion **1616B** around the second high-k dielectric layer **1514B**. In some embodiments, the first portion **1616A** of the WFA layer **1616** merges the second portion **1616B** of the WFA layer **1616** between the first high-k dielectric layer **1514A** and the second high-k dielectric layer **1514B**. In some embodiments, a distance between the first high-k dielectric layer **1414A** and the second high-k dielectric layer **1414B** is defined by the WFA layer **1616**. In some embodiments, the portion of the semiconductor structure **200** includes a first interfacial layer **1512A** between the first channel structure **1420A** and the first high-k dielectric layer **1514A**, and a second interfacial layer **1512B** between the second channel structure **1420B** and the second high-k dielectric layer **1514B**.

In some embodiments, the first portion **1616A** of the WFA layer **1616** is in direct contact with the second portion **1616B** of the WFA layer **1616**. In some embodiments, the first portion **1616A** of the WFA layer **1616** is in direct contact with the second portion **1616B** of the WFA layer **1616** so that the WFA layer **1616** has an hourglass shape between the channel structures **1420A** and **1420B**, in which a neck **1802** of the hourglass shape has a minimal width W_3 between the first high-k dielectric layer **1514A** and the second high-k dielectric layer **1514B**. In some embodiments, in a cross-section perpendicular to a top surface of the substrate **202** and perpendicular to a lengthwise direction of channel structures (i.e., a channel width cross-section as shown in FIG. 18), a width W_3 of the neck **1802** is larger than a width W_1 of the first channel structure **1420A** and a width W_2 of the second nanosheet structure **1420B**. In some embodiments, an interface **1804** is formed between the first portion **1616A** of the WFA layer **1616** and the second portion **1616B** of the WFA layer **1616**. In some embodiments, the interface **1804** includes a seam or a void. The seam or void may be vacuum due to the forming of the first WFA portion **1616A** and the second portion **1616B** under a substantially vacuum condition. In some embodiments, the first portion **1616A** of the WFA layer **1616** and the second portion **1616B** of the WFA layer **1616** are respectively a single-material or a uniformly formed layer comprising silicon element as shown in FIG. 18. In some embodiments, the first portion **1616A** of the WFA layer **1616** and the second portion **1616B** of the WFA layer **1616** are respectively a single layer of pure silicon. In some embodiments, due to the oxidation of the WFA layer **1616** as described above, an oxide layer **1616s**, such as a silicon oxide layer, is formed around the WFA layer **1616**.

In some embodiments, a thickness T_1 of the WFA layer **1616** is relatively small, e.g., less than about 15 Angstrom and the final gate structure **1400** with the WFA layer **1616** can still achieve a desired band-edge work function (e.g., about 4.2 eV) since the WFA layer **1616** contains silicon.

FIG. 19 illustrates a magnified cross-sectional view of a portion of the semiconductor structure **200** illustrated in FIG. 17C in accordance with some embodiments of the present disclosure. The portion of the semiconductor structure illustrated in FIG. 19 is similar to the semiconductor structure illustrated in FIG. 18, except that the first portion **1616A** and the second portion **1616B** of the WFA layer **1616** respectively have a double-layer structure. In some embodiments, the first portion **1616A** of the WFA layer **1616** includes an inner sub-WFA layer **1616A1** around the first high-k dielectric layer **1514A** and an outer sub-WFA layer **1616A2** around the inner sub-WFA layer **1616A1**. In some embodiments, the inner sub-WFA layer **1616A1** is a single

15

layer of pure silicon and the outer sub-WFA layer **1616A** is a single layer of transition metal silicide as described above. Similarly, in some embodiments, the second portion **1616B** of the WFA layer **1616** includes an inner sub-WFA layer **1616B1** around the second high-k dielectric layer **1514B** and an outer sub-WFA layer **1616B2** around the inner sub-WFA layer **1616B1**. In some embodiments, the inner sub-WFA layer **1616B1** is a single layer of pure silicon and the outer sub-WFA layer **1616B** is a single layer of transition metal silicide as described above. In some embodiments, the WFA layer **1616** has a triple-layer structure including an inner sub-WFA layer (such as the inner sub-WFA layer **1616A1** and the inner sub-WFA layer **1616B1**), an outer sub-WFA layer (such as the outer sub-WFA layer **1616A2** and the inner sub-WFA layer **1616B2**) around the inner sub-WFA layer, and an outermost sub-WFA layer (such as the oxide layer **1616s** as shown in FIG. **18**) around the outer sub-WFA layer. In some examples, the triple-layer structure of the WFA layer **1616** includes a pure silicon layer, a transition metal silicide layer around the pure silicon layer, and an oxide layer, such as a silicon oxide layer, around the transition metal silicide layer.

In some embodiments, an interface **1804** is formed between the outer sub-WFA layer **1616A2** and the outer sub-WFA layer **1616B2**. In some embodiments, the interface **1804** includes a seam or a void. The seam or void may be vacuum due to the forming of the outer sub-WFA layer **1616A2** and the outer sub-WFA layer **1616B2** in a substantially vacuum condition.

According to some aspects of the present disclosure, in some embodiments, a semiconductor structure includes a first gate structure around a first set of channel structures and a second gate structure around a second set of channel structures. In some embodiments, the first gate structure and the second gate structure are respectively as described and illustrated with respect to the final gate structure **1400** described above. In some embodiments, the first set of channel structures and the second set of channel structures are respectively as described and illustrated with respect to the channel structures **1420** described above. The first gate structure includes a first WFA layer, and the second gate structure includes a second WFA layer. In some embodiments, the first WFA layer and the second WFA layer are respectively differently arranged as described and illustrated with respect to the WFA layer **1616** described above with reference to FIG. **18**. In some embodiment, the first WFA layer includes pure silicon as described with reference to FIG. **18**, and the second WFA layer includes transition metal silicide as described with reference to FIG. **18**, where the pure silicon material of the WFA layer in the second gate structure is replaced by a transition metal silicide layer. A work function achieved by the first gate structure is lower than a work function achieved by the second gate structure. In some embodiments, the first gate structure can achieve a work function of around 4.2 eV, and the second gate structure can achieve a work function of around 4.3 eV. Therefore, the semiconductor structure allows for multiple threshold voltages (Vt). Alternatively, the first WFA layer includes pure silicon or metal silicide layer as described with reference to FIG. **18**, and the second WFA layer includes a double-layer structure as described with reference to FIG. **19**.

The present disclosure provides a semiconductor structure and a method for forming the semiconductor structure in accordance with some embodiments. The semiconductor structure includes a first nanosheet channel structure and a first high-k (HK) dielectric layer around the first nanosheet

16

channel structure. In addition, the semiconductor structure includes a second nanosheet channel structure disposed above the first nanosheet channel structure and a second high-k (HK) dielectric layer around the second nanosheet channel structure. The semiconductor structure further includes a Si-containing WFA layer between the first high-k dielectric layer and the second high-k dielectric layer. As a result, the semiconductor structure includes a “HK-Si-HK” structure between the first nanosheet channel structure and the second nanosheet channel structure. The WFA layer contains Si element, instead of Al element, and therefore can have a relatively small thickness (e.g., approximately 15 Angstrom or less) to achieve the desired band-edge work function (e.g., approximately 4.2 eV). Accordingly, the semiconductor structure in accordance with the present disclosure can be down-scaled and is suitable for use in advanced technology nodes.

In some embodiments, a semiconductor structure is provided. The semiconductor structure includes a first nanosheet channel structure, a first high-k dielectric layer surrounding the first nanosheet channel structure, a second nanosheet channel structure disposed above and substantially parallel to the first nanosheet channel structure, a second high-k dielectric layer surrounding the second nanosheet channel structure, and a work function adjustment layer including silicon and disposed between the first high-k dielectric layer and the second high-k dielectric layer. The first high-k dielectric layer and the second high-k dielectric layer are separated by the work function adjustment layer.

In some embodiments, a semiconductor structure is provided. The semiconductor structure includes a first nanosheet channel structure, and a first high-k dielectric layer surrounding the first nanosheet channel structure. In addition, the semiconductor structure includes a second nanosheet channel structure disposed above and substantially parallel to the first nanosheet channel structure, and a second high-k dielectric layer surrounding the second nanosheet channel structure. The semiconductor structure further includes a work function adjustment layer surrounding the first high-k dielectric layer and the second high-k dielectric layer. The work function adjustment layer includes an hourglass shape between the first high-k dielectric layer and the second high-k dielectric layer.

In some embodiments, a method for forming a semiconductor structure is provided. The method includes the following operations. A first nanosheet channel structure and a second nanosheet channel structure are formed. The second nanosheet channel structure is substantially parallel to the first nanosheet channel structure. A first high-k dielectric layer is formed all around the first nanosheet channel structure, and a second high-k dielectric layer is formed all around the second nanosheet channel structure. A silicon-based precursor is introduced to form a silicon-containing work function adjustment layer between the first high-k dielectric layer and the second high-k dielectric layer. A filling metal layer is formed to surround the first nanosheet channel structure, the second nanosheet channel structure, the first high-k dielectric layer, the second high-k dielectric layer and the silicon-containing work function adjustment layer.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments

17

introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure. 5

What is claimed is:

1. A semiconductor structure, comprising:
 - a first nanosheet channel structure;
 - a first high-k dielectric layer surrounding the first nanosheet channel structure; 10
 - a second nanosheet channel structure disposed above and substantially parallel to the first nanosheet channel structure;
 - a second high-k dielectric layer surrounding the second nanosheet channel structure; 15
 - a work function adjustment layer comprising silicon and disposed between the first high-k dielectric layer and the second high-k dielectric layer, wherein the work function adjustment layer comprises an elementary silicon portion in direct contact with the first and second high-k dielectric layers, and a silicon oxide coating in direct contact with the elementary silicon portion, wherein silicon elements of the silicon oxide coating are provided by the elementary silicon portion; 20
 - a metal layer in direct contact with the silicon oxide coating; and
 - a void disposed in the work function adjustment layer and between the first nanosheet channel structure and the second nanosheet channel structure, 25
 wherein the first high-k dielectric layer and the second high-k dielectric layer are separated by the work function adjustment layer, and the work function adjustment layer is substantially free of metal.
2. The semiconductor structure of claim 1, wherein the work function adjustment layer comprises a first portion surrounding the first high-k dielectric layer and a second portion surrounding the second high-k dielectric layer. 35
3. The semiconductor structure of claim 2, wherein the first portion of the work function adjustment layer merges the second portion of the work function adjustment layer between the first high-k dielectric layer and the second high-k dielectric layer. 40
4. The semiconductor structure of claim 1, wherein the silicon oxide coating is a native oxide layer on a surface of the elementary silicon portion. 45
5. The semiconductor structure of claim 1, wherein the work function adjustment layer has a thickness of no more than 15 angstrom.
6. The semiconductor structure of claim 1, further comprising: 50
 - a first interfacial layer between the first nanosheet channel structure and the first high-k dielectric layer, and a second interfacial layer between the second nanosheet channel structure and the second high-k dielectric layer; and 55
 - two spacer elements in contact with sidewalls of the first and second high-k dielectric layers and sidewalls of the first and second interfacial layers from a top-view perspective.
7. A semiconductor structure, comprising:
 - a first nanosheet channel structure;
 - a first high-k dielectric layer surrounding the first nanosheet channel structure;
 - a second nanosheet channel structure disposed above and substantially parallel to the first nanosheet channel structure; 65

18

- a second high-k dielectric layer surrounding the second nanosheet channel structure;
 - a work function adjustment layer comprising silicon and disposed around the first high-k dielectric layer and the second high-k dielectric layer, wherein the work function adjustment layer comprises an elementary silicon portion in direct contact with the first and second high-k dielectric layers, and a silicon oxide coating in direct contact with the elementary silicon portion, wherein silicon elements of the silicon oxide coating are provided by the elementary silicon portion;
 - a metal layer in direct contact with the silicon oxide coating; and
 - a void disposed in the work function adjustment layer and between the first nanosheet channel structure and the second nanosheet channel structure, 10
- wherein a portion of the work function adjustment layer is sandwiched between the first high-k dielectric layer and the second high-k dielectric layer, and the work function adjustment layer is substantially free of metal.
8. The semiconductor structure of claim 7, further comprising a first interfacial layer between the first nanosheet channel structure and the first high-k dielectric layer, and a second interfacial layer between the second nanosheet channel structure and the second high-k dielectric layer.
 9. A semiconductor structure, comprising:
 - a first nanosheet channel structure;
 - a first high-k dielectric layer surrounding the first nanosheet channel structure;
 - a second nanosheet channel structure disposed above and substantially parallel to the first nanosheet channel structure;
 - a second high-k dielectric layer surrounding the second nanosheet channel structure;
 - a work function adjustment layer disposed around the first high-k dielectric layer and the second high-k dielectric layer, wherein the work function adjustment layer comprises an elementary silicon portion in direct contact with the first and second high-k dielectric layers and a silicon oxide coating in direct contact with the elementary silicon portion, wherein silicon elements of the silicon oxide coating are provided by the elementary silicon portion;
 - a metal layer in direct contact with the silicon oxide coating; and
 - a void disposed in the work function adjustment layer and between the first nanosheet channel structure and the second nanosheet channel structure, 15
 wherein a portion of the work function adjustment layer extends from the first high-k dielectric layer to the second high-k dielectric layer, and the work function adjustment layer is substantially free of metal.
 10. The semiconductor structure of claim 9, further comprising a first interfacial layer between the first nanosheet channel structure and the first high-k dielectric layer, and a second interfacial layer between the second nanosheet channel structure and the second high-k dielectric layer.
 11. The semiconductor structure of claim 1, wherein from a top-view perspective, the first or second high-k dielectric layer includes a first portion and a second portion arranged on two sides of the work function adjustment layer, wherein each of the first portion and the second portion is between the work function adjustment layer and an inner spacer.
 12. The semiconductor structure of claim 1, wherein the first nanosheet channel structure and the second nanosheet channel structure have a rounded rectangular shape.

19

13. The semiconductor structure of claim 3, wherein the void is disposed at an interface between the first portion of the work function adjustment layer and the second portion of the work function adjustment layer.

14. The semiconductor structure of claim 7, wherein the silicon oxide coating is substantially free of metal.

15. The semiconductor structure of claim 7, wherein the first channel structure and the second channel structure have an oval shape.

16. The semiconductor structure of claim 9, further comprising a plurality of inner spacers having a rectangular shape and disposed on outer sides of the first and second high-k dielectric layers from a top-view perspective.

17. The semiconductor structure of claim 9, wherein the first nanosheet channel structure and the second nanosheet channel structure have a rounded rectangular shape.

18. The semiconductor structure of claim 13, wherein the void is vacuumed.

19. The semiconductor structure of claim 1, further comprising:

- a first dielectric layer laterally surrounding a first end of the first and second nanosheet channel structures;
- a second dielectric layer laterally surrounding a second end of the first and second nanosheet channel structures;
- a first spacer element covering the first dielectric layer, wherein the first dielectric layer and the first spacer element substantially have substantially identical widths from a top-view perspective; and
- a second spacer element covering the second dielectric layer, wherein the second dielectric layer and the

20

second spacer element substantially have substantially identical widths from a top-view perspective.

20. The semiconductor structure of claim 19, further comprises:

- a fin element protruding from a substrate, wherein the first nanosheet channel structure and the second nanosheet channel structure are disposed above the fin element;
- a plurality of isolation structures disposed on opposite sides of the fin;

source and drain features, wherein the first and second nanosheet channel structures extending between the source and drain features;

a plurality of first inner spacers each in contact with the source feature and one of the first and second high-k dielectric layers from a top-view perspective;

a plurality of second inner spacers each in contact with the drain feature and one of the first and second high-k dielectric layers from a top-view perspective; and

an inter-layer dielectric layer surrounding the source and drain features,

wherein the first spacer element separates the source feature from each of the first and second high-k dielectric layers, and separates the inter-layer dielectric layer from each of the first and second high-k dielectric layers; and

wherein the second spacer element separates the drain feature and each of the first and second high-k dielectric layers, and separates the inter-layer dielectric layer and each of the first and second high-k dielectric layers.

* * * * *